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# Chapter 1

## Introduction

Non-Newtonian and viscoelastic fluid flows are central to several industrial and biomedical applications, including polymer processing, extrusion, injection molding, and microfluidic lab-on-a-chip devices. Unlike Newtonian fluids, polymer solutions and macromolecular melts exhibit fading memory, non-linear stress-strain relationships, and elasticity. These characteristics give rise to distinctive hydrodynamic phenomena, such as rod climbing (the Weissenberg effect), die swell, and purely elastic flow instabilities that can develop even at vanishing Reynolds numbers where inertia is negligible.

Simulating these flows computationally remains an open challenge in continuum mechanics. Conventional discretization techniques, such as the Finite Element Method (FEM) and the Finite Volume Method (FVM), encounter the High Weissenberg Number Problem (HWNP). In this regime, hyperbolic stress advection generates steep localized gradients and thin boundary layers near stagnation points and solid walls, frequently leading to numerical instability and convergence breakdown. Furthermore, while established computational fluid dynamics packages handle forward simulations where constitutive parameters are known, identifying unknown rheological properties from kinematic observations requires computationally expensive iterative optimization loops.

Over the past decade, Scientific Machine Learning (SciML) and Physics-Informed Neural Networks (PINNs) have emerged as an alternative approach for continuum problems. By embedding the governing partial differential equations directly into the network loss through automatic differentiation, PINNs integrate forward flow reconstruction and inverse parameter estimation into a single continuous, mesh-free framework. While earlier benchmarks have focused on open-channel flows with net through-flow, the robustness, training stability, and identifiability limits of PINNs in enclosed recirculating geometries with mixed kinematics remain to be established.

This dissertation investigates the formulation, accuracy, and physical identifiability of physics-informed neural network architectures applied to the four-roll mill geometry, a configuration characterized by a central hyperbolic extensional stagnation point surrounded by shear layers adjacent to rotating cylinders.

## 1.1 Structure of the Dissertation

The remainder of this dissertation is organized into four sequential chapters:

**Chapter 2 (Viscoelastic Fluid Mechanics and the Four-Roll Mill):**

Establishes the continuum mechanics foundation, governing Navier–Stokes equations, and constitutive closures for Oldroyd-B and Giesekus fluids. It formally characterizes the four-roll mill cavity, its kinematic flow classification, and the stream-function parameterization.

**Chapter 3 (Physics-Informed Neural Networks in Complex Rheology):**

Reviews deep feedforward neural architectures, automatic differentiation, and the foundational PINN paradigm. It contrasts traditional machine learning generalization with continuous PDE boundary-value solving, and surveys prior literature benchmarks in computational rheology.

**Chapter 4 (Results and Discussion: Four-Roll Mill PINN Solver):**

Formulates the stream-function PINN solver for the four-roll mill and details the staged optimization strategy. It presents rigorous forward validations against high-fidelity COMSOL FEM solutions, inverse parameter estimation across linear and non-linear rheologies, grid independence analyses, transfer learning stability, and the mathematical origin of PTT shear flow degeneracies.

**Chapter 5 (Conclusions and Future Perspectives):** Synthesizes the core scientific findings of this research, reviews operational limitations regarding boundary stress supervision, and outlines promising future research directions.

## Chapter 2

# Viscoelastic Fluid Mechanics and the Four-Roll Mill

A polymer solution flowing through a four-roll mill behaves very differently from a Newtonian solvent. Macromolecular chains carry memory of their past deformation history, generating elastic stresses that persist across recirculation paths. In this chapter, we develop the continuum mechanics framework required to model these non-linear flows.

We begin from the universal conservation principles of mass and linear momentum, introducing the kinematic simplifications appropriate for low-inertia polymer solutions. We then formulate the constitutive equations governing viscoelastic fluids, starting from the linear Oldroyd-B model and the Upper-Convected Time Derivative (UCTD), and highlighting the kinematic origins of the High-Weissenberg Number Problem (HWNP). Next, we present the non-linear Giesekus and Phan-Thien–Tanner (PTT) formulations, which introduce stress-limiting mechanisms that bound extensional stress growth. Finally, we nondimensionalize the system using a viscous stress scale, characterize the four-roll mill geometry and its extensional stagnation kinematics, and define the forward and inverse problem formulations analyzed in this thesis.

## 2.1 General Conservation Laws

From a continuum perspective, fluid motion is governed by the universal conservation laws of mass and linear momentum [1].

### 2.1.1 Principle of Mass Conservation

For a fluid with local density  $\rho(\mathbf{x}, t)$  and velocity vector  $\mathbf{u}(\mathbf{x}, t)$ , conservation of mass is expressed by the continuity equation:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0. \quad (2.1)$$

Equation (2.1) states that the local rate of mass accumulation within a control volume balances the net convective mass flux through its boundaries.

### 2.1.2 Conservation of Linear Momentum

Applying Newton's second law to an infinitesimal fluid parcel yields the momentum balance in Eulerian coordinates:

$$\rho \left( \frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = \nabla \cdot \boldsymbol{\sigma} + \rho \mathbf{g}, \quad (2.2)$$

where  $\mathbf{g}$  represents external body force accelerations, and  $\boldsymbol{\sigma}$  is the symmetric second-order total Cauchy stress tensor. The left-hand side is the material derivative  $D\mathbf{u}/Dt$ , combining local unsteadiness  $\partial \mathbf{u}/\partial t$  with convective acceleration  $\mathbf{u} \cdot \nabla \mathbf{u}$ .

To isolate the isotropic pressure from deviatoric deformation stresses, we partition the total stress tensor  $\boldsymbol{\sigma}$  as:

$$\boldsymbol{\sigma} = -p\mathbf{I} + \mathbf{T}, \quad (2.3)$$

where  $p$  is the isotropic hydrodynamic pressure,  $\mathbf{I}$  is the second-order identity tensor, and  $\mathbf{T}$  is the extra-stress tensor containing all viscous and elastic contributions. Substituting Equation (2.3) into Equation (2.2) yields:

$$\rho \left( \frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla p + \nabla \cdot \mathbf{T} + \rho \mathbf{g}. \quad (2.4)$$

## 2.2 Physical Assumptions and Governing System Reduction

The full three-dimensional, unsteady equations of motion can be reduced by introducing scale-appropriate physical assumptions for polymer solutions within the four-roll mill geometry:

1. **Continuous medium and isotropic rest state:** The fluid is modeled as a continuous medium that remains locally isotropic at rest. Flow deformation stretches and aligns macromolecular chains, producing stress anisotropies represented by the extra-stress tensor  $\mathbf{T}$ .
2. **Incompressibility and stream function formulation:** Under the moderate operating pressures and low flow velocities considered here, density variations are negligible ( $\rho = \text{const}$ ). The continuity equation (2.1) simplifies to:

$$\nabla \cdot \mathbf{u} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0. \quad (2.5)$$

In two-dimensional planar motion, we satisfy incompressibility identically by introducing a scalar stream function  $\psi(x, y)$  such that:

$$u = \frac{\partial \psi}{\partial y}, \quad v = -\frac{\partial \psi}{\partial x}. \quad (2.6)$$

Because mixed partial derivatives commute ( $\partial^2 \psi / \partial x \partial y = \partial^2 \psi / \partial y \partial x$ ), the

velocity field satisfies  $\nabla \cdot \mathbf{u} \equiv 0$  everywhere.

3. **Isothermal operating conditions:** Internal temperature rises due to viscous heating remain negligible under our operating conditions. The energy equation decouples from the flow kinematics, and fluid transport properties (viscosities and relaxation times) are treated as constants.
4. **Absence of body forces:** The four-roll mill operates in a horizontal cross-sectional plane  $(x, y)$ . Gravitational acceleration acts along the vertical cylinder axis  $(z)$ , perpendicular to the flow plane, so we set  $\mathbf{g} = \mathbf{0}$ .
5. **Steady-state regime:** We focus on the established steady operating regime, setting all local temporal derivatives to zero ( $\partial(\cdot)/\partial t = \mathbf{0}$ ).
6. **Two-dimensional planar kinematics:** Given the high aspect ratio of the cylindrical rollers along the  $z$ -axis, axial end effects near the transverse mid-plane are negligible. The velocity field is modeled as strictly planar:

$$\mathbf{u} = (u(x, y), v(x, y), 0). \quad (2.7)$$

Under these physical simplifications, the steady momentum balance reduces to:

$$\rho(\mathbf{u} \cdot \nabla \mathbf{u}) = -\nabla p + \nabla \cdot \mathbf{T}. \quad (2.8)$$

## 2.3 Constitutive Modeling of Viscoelastic Fluids

In two dimensions, the momentum balance (2.8) contains six unknown scalar fields  $(u, v, p, T_{xx}, T_{xy}, T_{yy})$ , while mass and momentum conservation provide only

three equations. A constitutive equation for the extra-stress tensor  $\mathbf{T}$  is required to close the system.

For dilute and semi-dilute polymer solutions, we partition the extra-stress tensor  $\mathbf{T}$  into a Newtonian solvent contribution and an elastic polymeric contribution  $\boldsymbol{\tau}$  [1, 2]:

$$\mathbf{T} = 2\mu_s \mathbf{D} + \boldsymbol{\tau}, \quad (2.9)$$

where  $\mu_s$  is the dynamic viscosity of the Newtonian solvent, and  $\mathbf{D}$  is the symmetric rate-of-strain tensor:

$$\mathbf{D} = \frac{1}{2} [\nabla \mathbf{u} + (\nabla \mathbf{u})^T]. \quad (2.10)$$

This decomposition isolates the instantaneous viscous dissipation of the solvent ( $2\mu_s \mathbf{D}$ ) from the delayed elastic stress  $\boldsymbol{\tau}$  generated by macromolecular stretching and orientation. Substituting Equation (2.9) into Equation (2.8) and exploiting incompressibility ( $\nabla \cdot (2\mu_s \mathbf{D}) = \mu_s \nabla^2 \mathbf{u}$ ), the steady momentum equation becomes:

$$\rho(\mathbf{u} \cdot \nabla \mathbf{u}) = -\nabla p + \mu_s \nabla^2 \mathbf{u} + \nabla \cdot \boldsymbol{\tau}. \quad (2.11)$$

### 2.3.1 Upper-Convected Time Derivative

Polymeric fluids exhibit fading memory: macromolecules orient and stretch under flow and relax over finite timescales. As a fluid parcel moves through the domain, its internal stress is advected along streamlines, rotated by local vorticity, and stretched by velocity gradients.

To satisfy material frame-indifference, constitutive equations must employ an objective tensor rate. For polymer chains deforming affinely with the continuum, the appropriate objective rate is the *Upper-Convected Time Derivative* (UCTD),

denoted by  $\overset{\nabla}{\boldsymbol{\tau}}$ . With the velocity gradient defined as  $(\nabla \mathbf{u})_{ij} = \partial u_i / \partial x_j$ , the UCTD of the symmetric extra-stress tensor  $\boldsymbol{\tau}$  is:

$$\overset{\nabla}{\boldsymbol{\tau}} \equiv \frac{\partial \boldsymbol{\tau}}{\partial t} + \mathbf{u} \cdot \nabla \boldsymbol{\tau} - (\nabla \mathbf{u}) \boldsymbol{\tau} - \boldsymbol{\tau} (\nabla \mathbf{u})^T. \quad (2.12)$$

Under steady conditions,  $\partial \boldsymbol{\tau} / \partial t = \mathbf{0}$ . The term  $\mathbf{u} \cdot \nabla \boldsymbol{\tau}$  describes convective stress transport, while  $-(\nabla \mathbf{u}) \boldsymbol{\tau} - \boldsymbol{\tau} (\nabla \mathbf{u})^T$  captures stretching and rotation induced by local velocity gradients.

### 2.3.2 The Oldroyd-B Model and Extensional Behavior

The classical Oldroyd-B model [3] idealizes macromolecules as non-interacting Hookean dumbbells suspended in a Newtonian solvent [1]. Kinetic theory relates the polymeric viscosity  $\mu_p$  directly to the relaxation time  $\lambda$ :

$$\mu_p = nk_B T \lambda, \quad (2.13)$$

where  $n$  is polymer number density,  $k_B$  is the Boltzmann constant, and  $T$  is absolute temperature. Under steady conditions, the polymeric extra-stress tensor  $\boldsymbol{\tau}$  satisfies:

$$\boldsymbol{\tau} + \lambda \overset{\nabla}{\boldsymbol{\tau}} = 2\mu_p \mathbf{D}. \quad (2.14)$$

The total zero-shear viscosity of the solution is  $\mu_0 = \mu_s + \mu_p$ .

While Oldroyd-B predicts a constant shear viscosity  $\eta(\dot{\gamma}) = \mu_0$  (characteristic of Boger fluids), it exhibits a well-known singularity in steady planar extensional flow ( $u = Ex, v = -Ey$ , with  $E > 0$ ). As the strain rate approaches the critical threshold:

$$E_{\text{crit}} = \frac{1}{2\lambda}, \quad (2.15)$$

the assumption of infinitely extendable Hookean springs allows unbounded stretch-

ing. Consequently, extensional stresses and extensional viscosity diverge to infinity ( $\eta_E \rightarrow \infty$ ). At low deformation rates ( $E \rightarrow 0$ ), the planar extensional viscosity recovers the Newtonian limit, giving a planar Trouton ratio  $\text{Tr}_E = \eta_E/\mu_0 \rightarrow 4$ .

This stress singularity near  $2\lambda E \rightarrow 1$  is the constitutive root of the *High-Weissenberg Number Problem* (HWNP), which creates steep stress boundary layers that challenge conventional grid-based solvers [4].

### 2.3.3 Non-linear Closures: Giesekus and Phan-Thien–Tanner (PTT)

To avoid unbounded stress divergence and capture the behavior of concentrated solutions, non-linear constitutive models introduce stress-limiting mechanisms:

#### The Giesekus Model

The Giesekus model [1, 5] accounts for anisotropic hydrodynamic drag between neighboring macromolecules by adding a quadratic stress relaxation term:

$$\boldsymbol{\tau} + \lambda \overset{\nabla}{\boldsymbol{\tau}} + \frac{\alpha \lambda}{\mu_p} \boldsymbol{\tau}^2 = 2\mu_p \mathbf{D}, \quad (2.16)$$

where  $\alpha \in [0, 0.5]$  is the dimensionless mobility parameter, and the matrix square is  $(\boldsymbol{\tau}^2)_{ij} = \tau_{ik}\tau_{kj}$ . When  $\alpha = 0$ , the model recovers Oldroyd-B. For  $\alpha > 0$ , the quadratic term bounds extensional stress growth and introduces shear-thinning in steady shear.

#### The Phan-Thien–Tanner (PTT) Model

Derived from transient network theory [1, 6], the linear PTT model assumes that entanglement creation and destruction rates depend on stored elastic stress:

$$f(\boldsymbol{\tau})\boldsymbol{\tau} + \lambda \overset{\nabla}{\boldsymbol{\tau}} = 2\mu_p \mathbf{D}, \quad (2.17)$$

with the stress-softening function:

$$f(\boldsymbol{\tau}) = 1 + \frac{\varepsilon \lambda}{\mu_p} \text{tr}(\boldsymbol{\tau}), \quad (2.18)$$

where  $\text{tr}(\boldsymbol{\tau}) = \tau_{xx} + \tau_{yy}$ , and  $\varepsilon \geq 0$  is the extensibility parameter governing stress-induced entanglement breakdown. When  $\varepsilon = 0$ ,  $f = 1$  and the model recovers Oldroyd-B. For  $\varepsilon > 0$ , the relaxation rate increases with stress, bounding extensional viscosity.

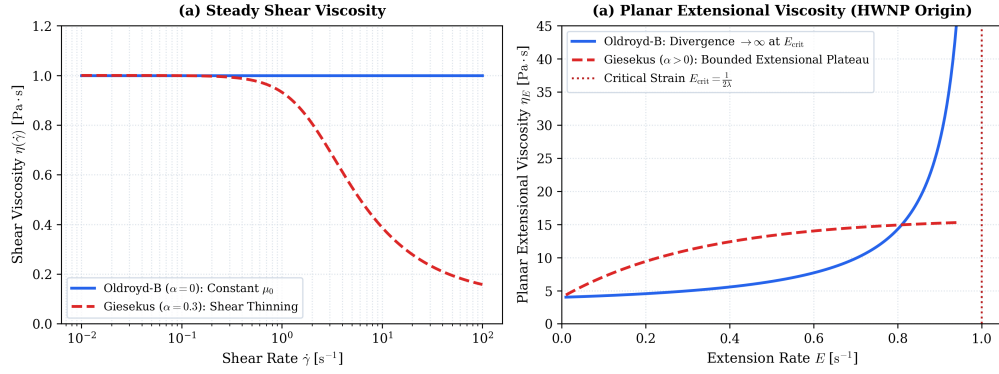


Figure 2.1: Conceptual comparison of constitutive model responses: (a) steady shear viscosity  $\eta(\dot{\gamma})$ , showing the constant viscosity of Oldroyd-B versus the shear-thinning behavior of Giesekus ( $\alpha > 0$ ); (b) planar extensional viscosity  $\eta_E(E)$ , contrasting the asymptotic stress divergence of Oldroyd-B at  $E_{\text{crit}} = 1/(2\lambda)$  with the bounded extensional plateau of the Giesekus formulation.

### Focus of Present Investigation: Oldroyd-B and Giesekus

In this dissertation, we focus our computational investigations on two representative models:

1. **Oldroyd-B:** The standard linear benchmark for constant-viscosity Boger fluids, capturing purely elastic normal stress growth and the High-Weissenberg Number Problem.
2. **Giesekus:** The primary non-linear model, incorporating mobility  $\alpha \in (0, 0.5]$  to capture shear-thinning and bounded extensional stress.

We keep the PTT formulation in our governing differential residuals for generality, setting it to  $\varepsilon = 0$  during baseline benchmarks and using it in Section 4.9.2 to study parameter identifiability limits.

## 2.4 Nondimensionalization and Governing Equations

Nondimensionalization scales physical variables to order unity and establishes the governing dimensionless groups [4]. In low-Reynolds viscoelastic flows, scaling by a viscous stress unit rather than an inertial dynamic pressure improves numerical conditioning and balances the magnitudes of the governing differential operators.

### 2.4.1 Reference Scales and Dimensionless Groups

We select reference length  $H_{\text{ref}}$  (roller radius  $R$ ), reference velocity  $U_{\text{ref}}$  (roller surface speed  $u_{\text{roll}}$ ), and a constant reference viscosity  $\eta_0$  (chosen as a fixed baseline scale, e.g.,  $\eta_0 = 1.0 \text{ Pa} \cdot \text{s}$ ). The dimensionless variables are defined as:

$$\mathbf{x}^* = \frac{\mathbf{x}}{H_{\text{ref}}}, \quad \mathbf{u}^* = \frac{\mathbf{u}}{U_{\text{ref}}}, \quad p^* = \frac{p}{\frac{\eta_0 U_{\text{ref}}}{H_{\text{ref}}}}, \quad \boldsymbol{\tau}^* = \frac{\boldsymbol{\tau}}{\frac{\eta_0 U_{\text{ref}}}{H_{\text{ref}}}}. \quad (2.19)$$

Dropping asterisks for notational brevity, the governing non-dimensional groups are:

$$Re_{\text{scale}} = \frac{\rho U_{\text{ref}} H_{\text{ref}}}{\eta_0} \quad (\text{Scaling Reynolds number}), \quad (2.20)$$

$$Re_{\text{phys}} = \frac{\rho U_{\text{ref}} H_{\text{ref}}}{\mu_s + \mu_p} = \frac{Re_{\text{scale}}}{\mu_s^* + \mu_p^*} \quad (\text{Physical Reynolds number}), \quad (2.21)$$

$$Wi = \frac{\lambda U_{\text{ref}}}{H_{\text{ref}}} \quad (\text{Global Weissenberg number}), \quad (2.22)$$

$$\beta = \frac{\mu_s}{\mu_s + \mu_p} = \frac{\mu_s^*}{\mu_s^* + \mu_p^*} \quad (\text{Viscosity ratio}), \quad (2.23)$$

$$\mu_s^* = \frac{\mu_s}{\eta_0}, \quad \mu_p^* = \frac{\mu_p}{\eta_0} \quad (\text{Dimensionless solvent and polymer viscosities}). \quad (2.24)$$

The distinction between  $Re_{\text{scale}}$  and  $Re_{\text{phys}}$  is essential in inverse problems. Defining the momentum equation through the constant scale  $Re_{\text{scale}}$  ensures that the numerical scaling of convective acceleration remains invariant even as unknown fluid viscosities  $(\mu_s, \mu_p)$  are adjusted during optimization.

### 2.4.2 Dimensionless Vector Formulation

The non-dimensional governing equations in steady 2D flow become:

$$\nabla \cdot \mathbf{u} = 0, \quad (2.25)$$

$$Re_{\text{scale}}(\mathbf{u} \cdot \nabla \mathbf{u}) + \nabla p - \mu_s^* \nabla^2 \mathbf{u} - \nabla \cdot \boldsymbol{\tau} = \mathbf{0}, \quad (2.26)$$

$$\mathcal{G}(\boldsymbol{\tau}) + Wi(\mathbf{u} \cdot \nabla \boldsymbol{\tau} - (\nabla \mathbf{u})\boldsymbol{\tau} - \boldsymbol{\tau}(\nabla \mathbf{u})^T) - 2\mu_p^* \mathbf{D} = \mathbf{0}, \quad (2.27)$$

where the constitutive operator  $\mathcal{G}(\boldsymbol{\tau})$  unifies the models under study:

$$\mathcal{G}(\boldsymbol{\tau}) = \begin{cases} \boldsymbol{\tau} & \text{(Oldroyd-B),} \\ \boldsymbol{\tau} + \frac{\alpha Wi}{\mu_p^*} \boldsymbol{\tau}^2 & \text{(Giesekus),} \\ \left(1 + \frac{\varepsilon Wi}{\mu_p^*} \text{tr}(\boldsymbol{\tau})\right) \boldsymbol{\tau} & \text{(Linear PTT).} \end{cases} \quad (2.28)$$

## 2.5 The Four-Roll Mill Benchmark Kinematics and Boundaries

The four-roll mill is a canonical benchmark designed to generate controlled two-dimensional planar extensional flow with a central stagnation point [7–10]. Originally conceived by Taylor [7] and systematically developed by Fuller and Leal [8, 9], it has served as a standard apparatus for analyzing macromolecular deformation and flow-induced birefringence. As depicted in Figure 2.2, the setup consists of four cylindrical rollers placed symmetrically inside a square domain.

### 2.5.1 Kinematic Relations and Flow Classification

In the vicinity of the central region between the rollers, we represent the local velocity field through a linear superposition of extension and rotation [8, 9]:

$$\mathbf{u}(x, y) = \begin{pmatrix} Ex + \omega y \\ -Ey - \omega x \end{pmatrix}, \quad (2.29)$$

where  $E \geq 0$  denotes the extension rate and  $\omega \geq 0$  represents the local angular rotation rate. The associated scalar vorticity is:

$$\omega_z = (\nabla \times \mathbf{u})_z = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = -2\omega. \quad (2.30)$$

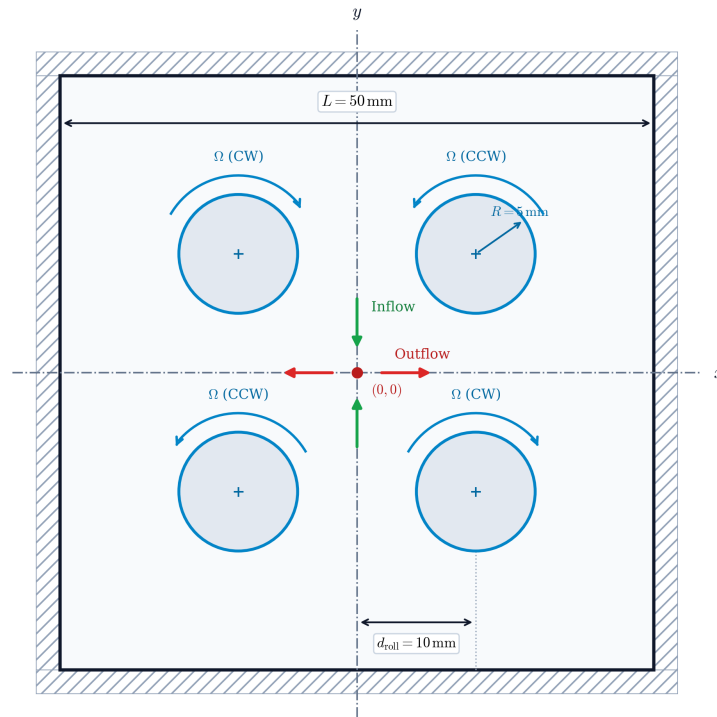


Figure 2.2: Schematic representation of the four-roll mill geometry, illustrating the square cavity of side  $L$ , cylindrical rollers of radius  $R$  at offset  $d_{\text{roll}}$ , Cartesian axes  $(x, y)$ , rotation directions with angular velocity  $\Omega$  (adjacent rolls counter-rotate, diagonal rolls co-rotate), and the central hyperbolic stagnation point  $(0, 0)$ .

To quantify the relative magnitude of extension versus rotation, we define a dimensionless flow type parameter  $\chi \in [-1, 1]$ :

$$\chi = \frac{E - \omega}{E + \omega}. \quad (2.31)$$

Depending on roller speeds and rotation directions, this idealized local kinematics captures three canonical flow regimes (illustrated in Figure 2.3):

$\chi = 1$  ( $\omega = 0$ ): **Pure Planar Extensional Flow.** Generated when adjacent rolls counter-rotate symmetrically (diagonal rolls co-rotate). Fluid elements experience pure stretching along  $x$  and pure compression along  $y$ , with hyperbolic streamlines  $xy = \text{const}$ .

$\chi = 0$  ( $E = \omega$ ): **Simple Shear Flow.** An intermediate state where rate of strain and angular rotation have equal magnitude ( $E = \omega$ ), producing rectilinear shear streamlines. In an enclosed cavity, this uniform shear field is realized locally near the center, while further away the flow curves to adapt to roller and wall boundaries.

$\chi = -1$  ( $E = 0$ ): **Pure Solid-Body Rotation.** Generated when rolls co-rotate to induce bulk circulation. This yields concentric circular streamlines around the origin, representing rigid-body rotation without elongation or compression.

### 2.5.2 The Central Hyperbolic Stagnation Point

In the extensional regime ( $\chi = 1$ ), the velocity vanishes at the geometric center:

$$\mathbf{u}(0, 0) = \mathbf{0}. \quad (2.32)$$

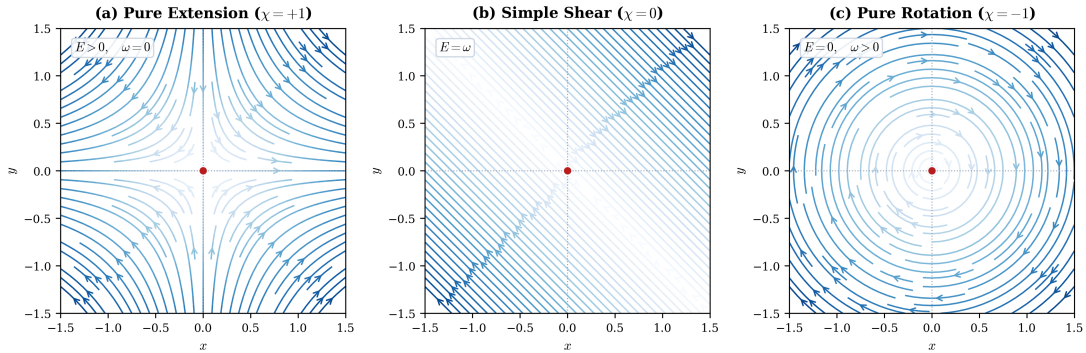


Figure 2.3: Streamline patterns of the two-dimensional linear flow field in Equation (2.29) for the three canonical regimes of the flow classification parameter  $\chi$ : (a) pure planar extension ( $\chi = +1$ ), (b) simple shear ( $\chi = 0$ ), and (c) pure solid-body rotation ( $\chi = -1$ ). The red marker indicates the central stagnation point  $(0, 0)$ .

The velocity gradient tensor at the origin,

$$\nabla \mathbf{u} = \begin{pmatrix} E & 0 \\ 0 & -E \end{pmatrix}, \quad (2.33)$$

possesses real eigenvalues  $+E$  (along the horizontal stretching axis  $x$ ) and  $-E$  (along the vertical compression axis  $y$ ), defining a hyperbolic saddle-point geometry. Along the incoming vertical streamline ( $x = 0$ ), fluid parcels decelerate toward zero velocity, providing fluid elements with long residence times.

This kinematic feature allows macromolecular deformation to accumulate over timescales comparable to or larger than the fluid relaxation time  $\lambda$ . Consequently, this setup provides an ideal testbed for observing elastic stress growth away from solid boundaries.

### 2.5.3 Geometric Setup and Operating Parameters

Following the established benchmark setup of D’Avino et al. [11], we define the physical configuration through:

**Cavity dimensions:** Square enclosure of side  $L = 50$  mm spanning  $[-25, 25]$  mm.

**Cylindrical rollers:** Radius  $R = 5 \text{ mm}$  ( $H_{\text{ref}} = R = 0.1L$ ).

**Roller placement:** Centers positioned symmetrically at  $(\pm 10, \pm 10) \text{ mm}$  ( $d_{\text{roll}} = 10 \text{ mm}$ ).

**Roller kinematics:** Peripheral surface velocity  $u_{\text{roll}} = \Omega R = 8.333 \times 10^{-3} \text{ m/s}$ .

**Non-dimensional domain:** In units of  $H_{\text{ref}} = R$ , the computational domain spans  $[-5, 5]^2$ , with roller centers located at  $(\pm 2.0, \pm 2.0)$  and non-dimensional roller radius  $R^* = 1.0$ .

The physical and operating parameters investigated in this work are summarized in Table 2.1.

Table 2.1: Physical and operating parameters for the four-roll mill benchmark.

| Parameter                        | Symbol               | Value / Range          | Unit              |
|----------------------------------|----------------------|------------------------|-------------------|
| Cavity side length               | $L$                  | 0.05                   | m                 |
| Roller radius (reference length) | $H_{\text{ref}} = R$ | 0.005                  | m                 |
| Roller surface velocity          | $u_{\text{roll}}$    | $8.333 \times 10^{-3}$ | m/s               |
| Fluid density                    | $\rho$               | 1000                   | kg/m <sup>3</sup> |
| Characteristic viscosity scale   | $\eta_0$             | 1.0                    | Pa · s            |
| Polymer relaxation time          | $\lambda$            | 0.05–1.0               | s                 |
| Viscosity ratio                  | $\beta$              | 0.1–0.5                | —                 |
| Scaling Reynolds number          | $Re_{\text{scale}}$  | $0.0417 \ll 1$         | —                 |
| Weissenberg number               | $Wi$                 | 0.083–1.667            | —                 |

## 2.5.4 Diagnostic Verification Cutlines

To enable rigorous, high-resolution validation of kinematic and stress profiles across the domain, we establish three representative diagnostic cutlines (Figure 2.4):

1. **Horizontal Cutline** ( $y = 0, -L/2 \leq x \leq L/2$ ): Traversing the central extensional outflow axis through the hyperbolic stagnation point  $(0, 0)$ . Along this path, fluid accelerates outward, forming intense elongational stress tails.

2. **Vertical Cutline** ( $x = 0, -L/2 \leq y \leq L/2$ ): Traversing the compressive inflow axis, capturing fluid deceleration into the central stagnation zone.
3. **Oblique Diagonal Cutline** ( $y = x$ ): Connecting opposite cavity corners ( $[-L/2, -L/2]$  to  $[L/2, L/2]$ ) and traversing the diagonal sectors. Because this trajectory intersects the solid cores of the top-right and bottom-left rollers (centered at  $(\pm 10, \pm 10)$  mm), spatial evaluations are sampled strictly within the active fluid domain. This captures the transition from the central extensional core into the high-gradient shear layers adjacent to roller surfaces.

These diagnostic trajectories serve as spatial evaluation baselines for both the reference CFD mesh convergence study and the PINN field validation detailed in Chapter 4.

### 2.5.5 Boundary Conditions and Pressure Point Anchoring

To complete the continuous boundary value formulation, we specify physical boundary conditions along all solid surfaces:

1. **Tangential Velocity on Roller Walls** ( $\Gamma_{\text{rolls}}$ ): On the surface of each roller  $k \in \{\text{TL}, \text{TR}, \text{BL}, \text{BR}\}$ , the fluid velocity satisfies no-slip with prescribed peripheral rotation:

$$\mathbf{u}|_{\Gamma_k} = \boldsymbol{\Omega}_k \times (\mathbf{x} - \mathbf{x}_k), \quad (2.34)$$

where  $\mathbf{x}_k$  is the roller center. For the planar extensional regime ( $\chi = 1$ ), adjacent rolls counter-rotate while diagonal rolls co-rotate: top-left and bottom-right rolls rotate clockwise ( $-\Omega$ ), while top-right and bottom-left rolls rotate counter-clockwise ( $+\Omega$ ).

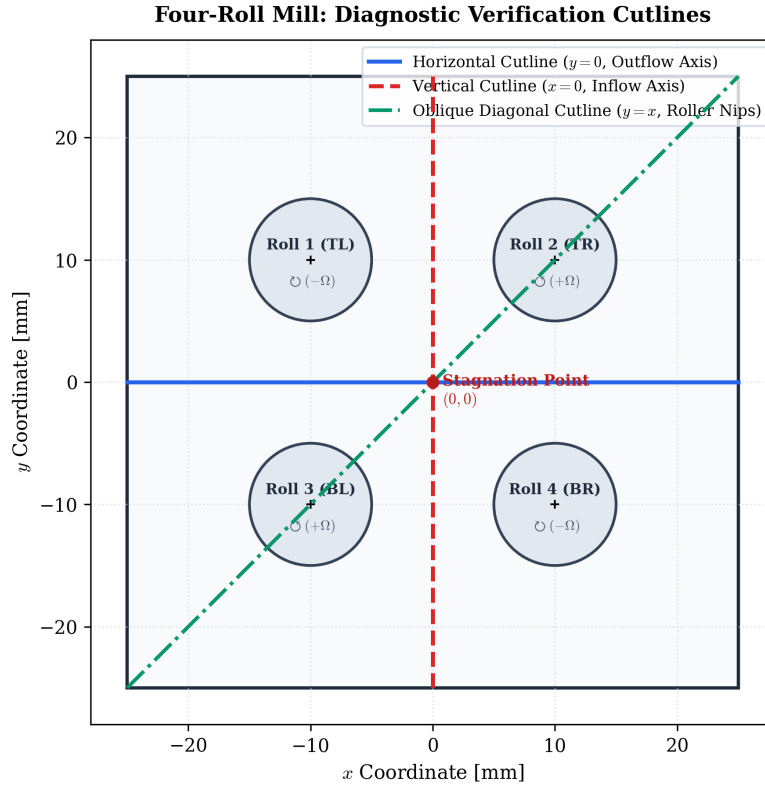


Figure 2.4: Schematic of the four-roll mill geometry depicting the four rotating cylinders (with adjacent rolls counter-rotating and diagonal rolls co-rotating to generate planar extensional flow), the central extensional stagnation point (0,0), and the three diagnostic verification cutlines: Horizontal ( $y = 0$ , blue), Vertical ( $x = 0$ , red dashed), and Oblique Diagonal ( $y = x$ , green dash-dot).

2. **No-Slip Velocity on Outer Cavity Walls** ( $\Gamma_{\text{outer}}$ ): Along the stationary external boundaries of the square enclosure, the velocity vanishes:

$$\mathbf{u}|_{\Gamma_{\text{outer}}} = \mathbf{0}. \quad (2.35)$$

3. **Pressure Point Anchoring**: Because pressure appears in the momentum balance (2.26) exclusively through its gradient  $\nabla p$ , in an enclosed cavity with prescribed velocity Dirichlet boundary conditions pressure is determined only up to an arbitrary additive constant ( $p \rightarrow p + C$ ). To fix this gauge freedom and obtain a unique solution, we specify a single reference pressure point:

$$p(\mathbf{x}_0) = 0, \quad (2.36)$$

at an arbitrary reference coordinate  $\mathbf{x}_0$  within the domain.

## 2.6 Problem Formulation and Computational Transition

The physical laws, constitutive closures, and boundary constraints established in this chapter define two complementary problem settings:

### 2.6.1 Forward and Inverse Problem Formulations

**Forward (Direct) Problem**: Given a complete set of known fluid parameters  $(\lambda, \mu_p, \mu_s, \alpha, \varepsilon)$  and prescribed kinematic boundary conditions on rolls and walls, determine the velocity field  $\mathbf{u}(x, y)$ , pressure field  $p(x, y)$ , and polymeric extra-stress tensor  $\boldsymbol{\tau}(x, y)$  that satisfy momentum balance (2.26) and constitutive closure (2.27) throughout cavity domain  $\Omega$ .

**Inverse (Parameter Identification) Problem:** Given partial kinematic observations of the velocity field within the cavity, reconstruct the latent internal fields  $(p, \boldsymbol{\tau})$  and identify the unknown physical parameters governing the fluid  $(\lambda, \mu_p, \mu_s, \text{ alongside non-linear parameters } \alpha \text{ or } \varepsilon)$ .

Table 2.2 classifies the role of each variable and parameter across the two problem formulations.

Table 2.2: Classification of physical fields and parameters across forward and inverse problem formulations.

| Physical Quantity   | Symbol                                                  | Forward Problem               | Inverse Problem                            |
|---------------------|---------------------------------------------------------|-------------------------------|--------------------------------------------|
| Velocity Field      | $\mathbf{u} = (u, v)$                                   | Primary unknown (via $\psi$ ) | Observed / Supervised                      |
| Pressure Field      | $p$                                                     | Primary unknown               | Latent (unobserved)                        |
| Polymeric Stress    | $\boldsymbol{\tau} = (\tau_{xx}, \tau_{xy}, \tau_{yy})$ | Primary unknown               | Latent (unobserved)                        |
| Relaxation Time     | $\lambda$ ( $Wi$ )                                      | Known parameter               | <b>Unknown physical parameter</b>          |
| Polymeric Viscosity | $\mu_p$                                                 | Known parameter               | <b>Unknown physical parameter</b>          |
| Solvent Viscosity   | $\mu_s$                                                 | Known parameter               | <b>Unknown physical parameter</b>          |
| Giesekus Mobility   | $\alpha$                                                | Known parameter               | <b>Unknown parameter / Model selection</b> |
| PTT Extensibility   | $\varepsilon$                                           | Known parameter               | <b>Unknown parameter / Model selection</b> |

## 2.6.2 Transition to Physics-Informed Neural Networks

The differential system formulated in Equations (2.25)–(2.28) represents a highly coupled, non-linear system of partial differential equations. In Chapter 3, we translate this continuous physical formulation into a machine-learning framework using Physics-Informed Neural Networks (PINNs). By embedding momentum and constitutive equations directly into the neural network loss function through automatic differentiation, the PINN framework enables both forward flow reconstruction and inverse rheological parameter discovery without requiring traditional spatial discretization grids.

## Chapter 3

# Physics-Informed Neural Networks in Complex Rheology

Feedforward deep neural networks, commonly known as Multi-Layer Perceptrons (MLPs), provide flexible continuous approximations for non-linear functions [12]. An MLP is formed by composing a sequence of parametric functions, termed *layers*. Each layer maps its inputs through a linear transformation followed by an element-wise non-linear operation. By cascading these layers in series, the network can represent continuous functional mappings of arbitrary complexity without requiring hand-crafted basis functions.

In scientific computing, neural networks offer an appealing feature: they represent fields as continuous, globally defined functions over spatial coordinates. Rather than storing state variables at discrete grid vertices, the network parameterizes physical fields everywhere in the continuum. However, using purely data-driven neural networks for fluid mechanics presents fundamental challenges: they require large amounts of labeled data, can violate conservation laws, and cannot infer latent material properties on their own.

Physics-Informed Neural Networks (PINNs) address these challenges by embedding governing partial differential equations directly into the training ob-

jective. In this chapter, we establish the mathematical foundations of neural networks and their optimization, discuss why continuous neural surrogates are suitable for transport phenomena, formulate the PINN paradigm, and review foundational benchmarks in complex fluid mechanics.

## 3.1 Fundamentals of Deep Learning

Deep learning models approximate continuous mathematical mappings by learning hierarchical representations directly from data [12]. In this section, we review the fundamental building blocks of feedforward networks: the artificial neuron, activation functions, multi-layer architectures, loss functions, optimization algorithms, and gradient computation.

### 3.1.1 The Artificial Neuron and Non-Linear Activation Functions

The elementary computational unit of a neural network is the artificial neuron. Given an input vector  $\mathbf{x} \in \mathbb{R}^d$ , a neuron computes an affine combination of the inputs followed by a non-linear activation:

$$z = \mathbf{w} \cdot \mathbf{x} + b, \quad a = \sigma(z), \quad (3.1)$$

where  $\mathbf{w}$  is the weight vector,  $b$  is a scalar bias, and  $\sigma(\cdot)$  is the activation function. Non-linear activations are essential: without them, composing multiple linear layers collapses algebraically into a single linear mapping, depriving deep architectures of their expressive power.

In classical machine learning, piecewise-linear functions such as the Rectified Linear Unit,  $\text{ReLU}(z) = \max(0, z)$ , are widely favored for their computational efficiency. However, in scientific computing and fluid mechanics, the network

outputs parameterize continuous physical fields governed by differential equations. These governing balance laws involve spatial derivatives of first, second, or third order, such as viscous diffusion  $\mu_s \nabla^2 \mathbf{u}$  or stress divergence  $\nabla \cdot \boldsymbol{\tau}$ .

Because the first derivative of ReLU is discontinuous and its second derivative is zero almost everywhere, evaluating a ReLU network inside differential residuals eliminates diffusive and curvature terms. Physics-informed models therefore require smooth, differentiable activation functions ( $\mathcal{C}^2$  or higher). In this work, we employ the Sigmoid Linear Unit (SiLU, also known as Swish) [12]:

$$\text{SiLU}(z) = \frac{z}{1 + e^{-z}}. \quad (3.2)$$

The SiLU activation is smooth, non-monotonic, and self-gated. Unlike the hyperbolic tangent ( $\tanh$ ), which saturates toward constant values for large positive inputs, SiLU remains active in the positive regime. This property maintains healthy gradient propagation in flow regions with intense shear and steep localized stress gradients, which are characteristic of viscoelastic flows.

### 3.1.2 Multi-Layer Perceptron Architecture

By arranging artificial neurons into sequential layers, we construct a Multi-Layer Perceptron (MLP). The network maps an input vector  $\mathbf{x}$  to an output prediction  $\hat{\mathbf{u}}$  by propagating information through  $L$  successive layers:

$$\begin{aligned} \mathbf{h}^{(1)} &= \sigma \left( \mathbf{W}^{(1)} \mathbf{x} + \mathbf{b}^{(1)} \right), \\ \mathbf{h}^{(l)} &= \sigma \left( \mathbf{W}^{(l)} \mathbf{h}^{(l-1)} + \mathbf{b}^{(l)} \right), \quad \text{for } l = 2, \dots, L-1, \\ \hat{\mathbf{u}} &= \mathbf{W}^{(L)} \mathbf{h}^{(L-1)} + \mathbf{b}^{(L)}, \end{aligned} \quad (3.3)$$

where  $\mathbf{h}^{(l)}$  denotes the hidden feature vector at layer  $l$ , and  $\mathbf{W}^{(l)}$  and  $\mathbf{b}^{(l)}$  represent the weight matrix and bias vector of that layer. In our implementation, the inputs

$\mathbf{x} = (x, y)$  are spatial coordinates within the fluid cavity, while the outputs  $\hat{\mathbf{u}}$  parameterize the physical state variables: the stream function  $\psi$ , hydrodynamic pressure  $p$ , and extra-stress tensor components  $\boldsymbol{\tau}$ .

A central theoretical property of MLPs is universal function approximation [13,14]: a feedforward network with sufficient width and smooth non-linear activations can approximate any continuous function on a compact domain to arbitrary precision. However, universal approximation guarantees only the existence of a suitable set of weights, not that standard gradient-based optimization will automatically find them or that spatial derivatives will match the physical solution with equal fidelity.

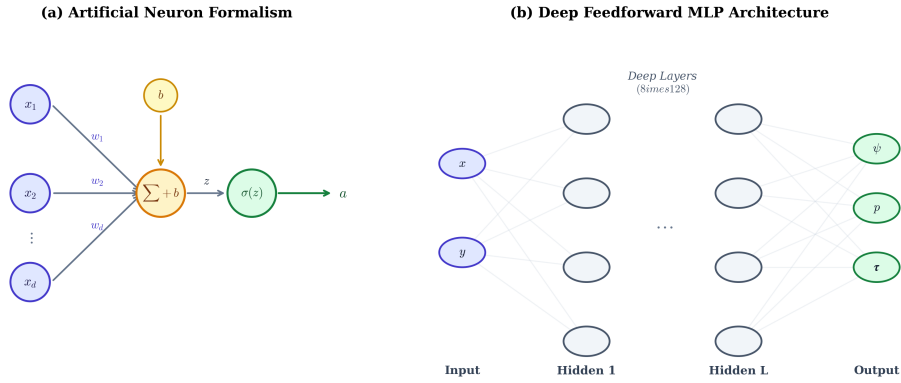


Figure 3.1: Neural network representations: (a) schematic of an artificial neuron computing an affine combination  $z = \mathbf{w} \cdot \mathbf{x} + b$  followed by a smooth activation  $\sigma(z)$ ; (b) deep Multi-Layer Perceptron (MLP) architecture propagating coordinate inputs  $(x, y)$  through  $L$  hidden layers to parameterize continuous physical fields  $(\psi, p, \boldsymbol{\tau})$ .

### Weight Initialization: Glorot Scheme and Output Zero-Initialization

The initialization of network weights strongly influences optimization dynamics. To prevent vanishing or exploding activations across deep layers, hidden layer weights are initialized using the Xavier/Glorot normal distribution [12], which scales parameter variances inversely with layer widths.

For the final linear layers predicting the extra-stress tensor  $\boldsymbol{\tau}$  and pressure

$p$ , we initialize weights and biases to zero ( $\mathbf{W}^{(L)} = \mathbf{0}, \mathbf{b}^{(L)} = \mathbf{0}$ ). This choice provides two practical benefits:

1. **Physical state of rest:** In fluid mechanics, a fluid at rest carries zero extra-stress ( $\boldsymbol{\tau} = \mathbf{0}$ ). Zero-initialization ensures that at iteration zero, the stress network starts from an unperturbed resting state.
2. **Suppression of initial gradient shocks:** If output weights were initialized randomly, initial stress predictions would contain high-frequency spatial fluctuations. Evaluating these fluctuations inside non-linear constitutive equations, which involve quadratic stress products and convective gradients, would generate large initial residual spikes that can destabilize optimization. Zero-initializing the output layer ensures bounded, well-behaved initial residuals.

### 3.1.3 Loss Functions and Optimization Algorithms

Training an MLP consists of finding parameters  $\boldsymbol{\theta} = \{\mathbf{W}^{(l)}, \mathbf{b}^{(l)}\}$  that minimize a scalar objective function  $\mathcal{L}(\boldsymbol{\theta})$ . In supervised regression, the discrepancy between predictions and reference values is commonly evaluated using the Mean Squared Error (MSE):

$$\mathcal{L}_{\text{MSE}}(\boldsymbol{\theta}) = \frac{1}{N} \sum_{i=1}^N \|\hat{\mathbf{u}}(\mathbf{x}_i; \boldsymbol{\theta}) - \mathbf{u}_i^{\text{target}}\|_2^2. \quad (3.4)$$

Parameters are updated iteratively along the negative gradient:

$$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta \nabla_{\boldsymbol{\theta}} \mathcal{L}, \quad (3.5)$$

where  $\eta > 0$  is the learning rate governing the step size.

In scientific PINN applications, optimization commonly employs a two-tier strategy:

1. **First-order adaptive optimization (Adam):** The Adaptive Moment Estimation algorithm [15] scales parameter updates using running estimates of past gradient moments. This enables Adam to traverse non-convex loss landscapes effectively during early exploration.
  
2. **Second-order quasi-Newton optimization (L-BFGS):** Once the parameters reach the basin of attraction of a minimum, the Limited-memory Broyden–Fletcher–Goldfarb–Shanno (L-BFGS) algorithm incorporates curvature approximations to achieve rapid, deterministic convergence, driving PDE residuals to low tolerance levels.

### 3.1.4 Backpropagation and Computational Graphs

To update network parameters via gradient descent, we compute the partial derivatives of the scalar loss with respect to all weights and biases,  $\nabla_{\theta}\mathcal{L}$ . Evaluating these gradients through individual numerical perturbations would be computationally prohibitive.

Instead, neural networks compute analytical gradients using the *backpropagation* algorithm [12]. The forward evaluation is structured as a directed acyclic graph of mathematical operations. Backpropagation applies the calculus chain rule in reverse order, starting from the loss scalar and propagating sensitivities backward through the graph. This reverse-mode automatic differentiation evaluates all parameter gradients simultaneously at a computational cost comparable to a single forward pass, making deep network training feasible.

## 3.2 Why Neural Networks? Capabilities and Physical Limitations

### 3.2.1 Capabilities and Universal Function Approximation

Neural networks offer distinctive characteristics that make them attractive for continuous physical modeling:

1. **Continuous and Meshless Representation:** Traditional grid-based methods (Finite Element, Finite Volume, and Finite Difference Methods) represent state variables exclusively at discrete vertices or cell centers, relying on explicit mesh connectivity. In contrast, a neural network provides a continuous, globally defined mapping  $\mathbf{x} \mapsto \hat{\mathbf{u}}(\mathbf{x})$  over the entire domain, capable of being evaluated at arbitrary spatial coordinates. However, being meshless does not imply freedom from geometric considerations: defining the domain, enforcing boundary conditions, and designing effective collocation point distributions remain essential and influential steps.
2. **High-Dimensional Functional Approximation:** The universal approximation capacity allows neural networks to represent complex, coupled physical fields (velocities, pressures, and stress tensors) within a unified functional representation, without requiring specialized interpolation elements for each variable.
3. **Hardware Acceleration and Differentiability:** Modern deep learning frameworks are built upon dense linear algebra operations optimized for parallel accelerator hardware (GPUs and TPUs), and provide built-in automatic differentiation for computing spatial derivatives of the represented fields.

### 3.2.2 Critical Bottlenecks of Purely Data-Driven Machine Learning

Despite their high expressivity, deploying purely data-driven black-box networks to fluid mechanics and non-Newtonian rheology reveals fundamental limitations:

**Data scarcity in experimental rheology:** Deep neural networks typically rely on large labeled datasets. In fluid experiments, acquiring dense velocity fields via Particle Image Velocimetry (PIV) is labor-intensive, while non-intrusive measurements of internal extra-stress components are generally impossible in complex geometries.

**Violation of fundamental conservation laws:** Purely data-driven models minimize empirical loss on discrete observations without physical constraints. Consequently, their predictions routinely violate incompressibility ( $\nabla \cdot \mathbf{u} \neq 0$ ), momentum balance, and constitutive laws, producing non-physical artifacts when interpolating or extrapolating.

**Overfitting and spurious oscillations:** In sparse-data regimes, overparameterized networks can easily fit measurement noise, yielding low training errors while developing spurious high-frequency oscillations in unmonitored regions.

**Inability to discover governing parameters:** Black-box models learn input-output mappings, but cannot extract underlying physical properties—such as relaxation times or polymeric viscosities—that govern flow dynamics.

These limitations show that data alone are insufficient for scientific modeling in rheology, motivating frameworks that embed physical conservation laws directly into the loss function.

### 3.3 The Physics-Informed Neural Network Paradigm

The Physics-Informed Neural Network (PINN) framework, formalized by Raissi, Perdikaris, and Karniadakis [16], addresses the shortcomings of purely data-driven modeling by incorporating differential governing equations directly into the network training objective.

#### 3.3.1 Physics-Constrained Loss Formulation

Consider a general non-linear differential operator  $\mathcal{F}$  governing a physical state  $\mathbf{u}(\mathbf{x})$  over a spatial domain  $\Omega \subset \mathbb{R}^d$  bounded by  $\partial\Omega$ :

$$\mathcal{F}[\mathbf{u}(\mathbf{x}); \boldsymbol{\lambda}] = \mathbf{0}, \quad \mathbf{x} \in \Omega, \quad (3.6)$$

$$\mathcal{B}[\mathbf{u}(\mathbf{x})] = \mathbf{g}(\mathbf{x}), \quad \mathbf{x} \in \partial\Omega, \quad (3.7)$$

where  $\boldsymbol{\lambda}$  represents physical parameters (such as viscosities or relaxation times) and  $\mathcal{B}$  denotes boundary operators.

In a PINN, the unknown physical fields  $\mathbf{u}(\mathbf{x})$  are parameterized by a neural network  $\hat{\mathbf{u}}(\mathbf{x}; \boldsymbol{\theta})$ . Evaluating the governing equations on the network outputs defines the differential residual:

$$\mathbf{f}(\mathbf{x}; \boldsymbol{\theta}, \boldsymbol{\lambda}) \equiv \mathcal{F}[\hat{\mathbf{u}}(\mathbf{x}; \boldsymbol{\theta}); \boldsymbol{\lambda}]. \quad (3.8)$$

The network weights  $\boldsymbol{\theta}$ , alongside unknown physical parameters  $\boldsymbol{\lambda}$  in inverse problems, are optimized by minimizing a composite multi-objective loss:

$$\mathcal{L}_{\text{total}}(\boldsymbol{\theta}, \boldsymbol{\lambda}) = w_{\text{data}}\mathcal{L}_{\text{data}} + w_{\text{bc}}\mathcal{L}_{\text{bc}} + w_{\text{pde}}\mathcal{L}_{\text{pde}}, \quad (3.9)$$

where:

$$\mathcal{L}_{\text{data}} = \frac{1}{N_{\text{data}}} \sum_{i=1}^{N_{\text{data}}} \left\| \hat{\mathbf{u}}(\mathbf{x}_i^{\text{data}}) - \mathbf{u}_i^{\text{data}} \right\|_2^2, \quad (3.10)$$

$$\mathcal{L}_{\text{bc}} = \frac{1}{N_{\text{bc}}} \sum_{j=1}^{N_{\text{bc}}} \left\| \mathcal{B}[\hat{\mathbf{u}}(\mathbf{x}_j^{\text{bc}})] - \mathbf{g}(\mathbf{x}_j^{\text{bc}}) \right\|_2^2, \quad (3.11)$$

$$\mathcal{L}_{\text{pde}} = \frac{1}{N_{\text{pde}}} \sum_{k=1}^{N_{\text{pde}}} \left\| \mathbf{f}(\mathbf{x}_k^{\text{pde}}; \boldsymbol{\theta}, \boldsymbol{\lambda}) \right\|_2^2. \quad (3.12)$$

Here,  $\{\mathbf{x}_i^{\text{data}}\}$  denotes available empirical measurements,  $\{\mathbf{x}_j^{\text{bc}}\}$  represents points sampled along physical boundaries  $\partial\Omega$ , and  $\{\mathbf{x}_k^{\text{pde}}\}$  designates interior *collocation points* scattered across  $\Omega$ .

This formulation naturally supports two complementary problem settings:

**Forward problems:** The physical parameters  $\boldsymbol{\lambda}$  are known. The network optimizes its weights  $\boldsymbol{\theta}$  to solve for the continuous fields  $\hat{\mathbf{u}}(\mathbf{x})$ , acting as a meshless continuous PDE solver.

**Inverse problems:** One or more constitutive parameters in  $\boldsymbol{\lambda}$  are unknown. The network simultaneously reconstructs the unobserved latent fields and identifies the unknown parameters by combining empirical velocity observations  $\mathcal{L}_{\text{data}}$  with physical consistency constraints  $\mathcal{L}_{\text{pde}}$ .

A key methodological principle of this thesis is the distinction between field reconstruction accuracy and parameter identifiability. A network may fit observed velocity fields accurately, yet inferred rheological parameters may remain uncertain if the kinematics lack sufficient variation or if the loss landscape exhibits degeneracies. Embedding physical equations provides essential structure, but does not automatically guarantee unique parameter recovery.

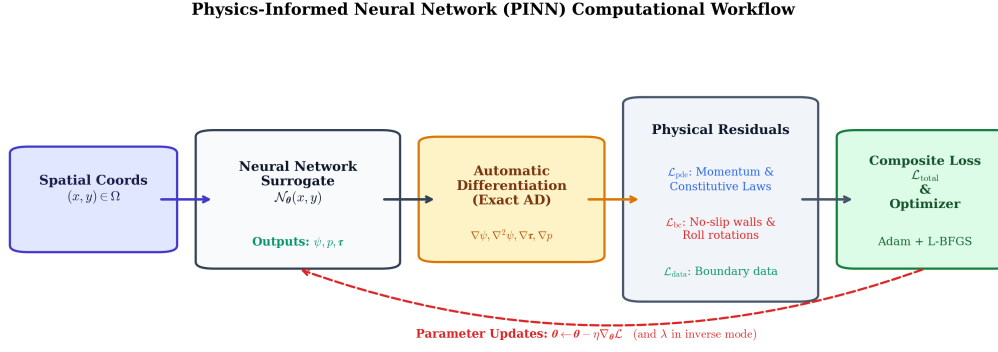


Figure 3.2: The Physics-Informed Neural Network (PINN) architecture: coordinate inputs  $(x, y)$  are mapped through a deep surrogate, spatial derivatives are evaluated via Automatic Differentiation (AD), and weights  $\theta$  alongside unknown physical parameters  $\lambda$  are updated iteratively by minimizing a composite loss combining PDE residuals, boundary conditions, and observed data.

### 3.3.2 Standard Supervised Training vs. PINN Optimization

Several operational features distinguish physics-informed training from conventional supervised learning:

1. **Training as a boundary-value solver:** In classical machine learning, a model is trained on a sample set to generalize to unseen statistical distributions. In PINNs, *training is the numerical solver*. The network is not designed to predict flows across arbitrary unseen geometries; rather, optimization directly solves the boundary-value problem on the specified domain  $\Omega$ . Available measurements  $\mathcal{L}_{\text{data}}$  act as an active data-assimilation anchor guiding the network toward the true physical state.
2. **Physical regularization and inductive bias:** In standard supervised training, the network is unconstrained between data points. In PINNs, evaluating differential residuals  $\mathcal{L}_{\text{pde}}$  across interior collocation points penalizes non-physical gradients and curvatures. This analytical inductive

bias restricts the admissible solution space. However, proper weighting between data and PDE terms remains essential: poorly tuned weights can still bias the optimization or slow down convergence.

**3. Coordinate Automatic Differentiation (AD):** Classical deep learning uses backpropagation to differentiate the loss with respect to network parameters:

$$\text{Backpropagation (Classical DL): } \frac{\partial \mathcal{L}}{\partial \boldsymbol{\theta}}. \quad (3.13)$$

In PINNs, the same reverse-mode automatic differentiation engine computes spatial derivatives of the predicted state fields with respect to input spatial coordinates:

$$\text{Coordinate AD (PINNs): } \frac{\partial \hat{\mathbf{u}}}{\partial \mathbf{x}}, \quad \frac{\partial^2 \hat{\mathbf{u}}}{\partial \mathbf{x}^2}. \quad (3.14)$$

As summarized in Table 3.1, coordinate AD evaluates exact derivatives of the neural functional representation along the computational graph, avoiding the spatial truncation error  $\mathcal{O}(\Delta x^2)$  of finite-difference stencils. The accuracy of these derivatives with respect to the true physical field depends on how closely the network itself approximates the physical solution.

**4. Collocation point sampling:** Unlike traditional CFD solvers that require conforming meshes and element connectivity tables, PINNs evaluate differential residuals at scattered collocation points  $\{\mathbf{x}_k^{\text{pde}}\}$ . This meshless feature simplifies domain representation, though adequate point density remains necessary in boundary layers and near stagnation regions.

Table 3.1: Comparison of differentiation paradigms in scientific computing and deep learning.

| Method                                | Formulation                                       | Truncation Error                | Grid Dependence              |
|---------------------------------------|---------------------------------------------------|---------------------------------|------------------------------|
| Finite Differences (FDM)              | $\frac{u(x+\Delta x) - u(x-\Delta x)}{2\Delta x}$ | $\mathcal{O}(\Delta x^2)$       | Structured Mesh Stencil      |
| Symbolic Differentiation              | Closed-form analytical expression                 | None                            | Expression Growth            |
| <b>Automatic Differentiation (AD)</b> | Exact chain rule on computational graph           | <b>None (Machine Precision)</b> | <b>Meshless / Continuous</b> |

## 3.4 Foundational Literature Benchmarks

### 3.4.1 Navier–Stokes Flow Past a Cylinder (Raissi et al., 2019)

In their foundational paper, Raissi, Perdikaris, and Karniadakis [16] demonstrated the capability of PINNs on laminar vortex shedding past a circular cylinder at  $Re = 100$ .

#### Problem Formulation and Inverse Discovery

The non-dimensional incompressible Navier–Stokes equations were formulated as:

$$u_t + \lambda_1(uu_x + vv_y) = -p_x + \lambda_2(u_{xx} + u_{yy}), \quad (3.15)$$

$$v_t + \lambda_1(uv_x + vv_y) = -p_y + \lambda_2(v_{xx} + v_{yy}), \quad (3.16)$$

$$u_x + v_y = 0, \quad (3.17)$$

where  $\lambda_1 = 1.0$  governs convective acceleration and  $\lambda_2 = 1/Re = 0.01$  represents kinematic viscosity.

To satisfy incompressibility identically, the authors parameterized velocities via a stream function  $\psi(t, x, y)$  ( $u = \partial\psi/\partial y, v = -\partial\psi/\partial x$ ). A neural network with 9 hidden layers and 20 neurons per layer, activated by tanh, predicted  $(\psi, p)$  while treating  $\lambda_1$  and  $\lambda_2$  as learnable parameters.

#### Latent Field Discovery and Parameter Identification

The network was trained on  $N = 5,000$  spatio-temporal velocity observations in the cylinder wake. The benchmark demonstrated two key capabilities:

**Latent pressure recovery:** Without any pressure measurements, the network reconstructed the pressure field  $p(t, x, y)$  solely by enforcing the mo-

momentum balance, matching CFD reference data up to an arbitrary constant.

**Parameter estimation under noise:** On clean data, parameter errors were 0.08% for  $\lambda_1$  and 4.67% for  $\lambda_2$ . Under 1% uncorrelated Gaussian noise, errors remained controlled at 0.17% and 5.70%.

While this benchmark validated simultaneous field reconstruction and parameter identification, it was restricted to linear Newtonian fluids in an open domain.

### 3.4.2 The ViscoelasticNet Framework (Thakur et al., 2024)

Extending physics-informed methods to non-Newtonian media, Thakur, Raissi, and Ardekani [17] introduced *ViscoelasticNet*, applying PINNs to polymeric flows governed by differential constitutive equations.

#### The Rheological Bottleneck

Viscoelastic fluid simulation presents two major challenges:

1. **Experimental limitations:** Non-intrusive measurements of internal extra-stresses  $\boldsymbol{\tau}$  are generally unavailable in complex geometries.
2. **Numerical instability (HWNP):** In standard grid-based solvers (FEM, FVM), elevated elasticity triggers the High-Weissenberg Number Problem (HWNP), where steep stress boundary layers cause convergence breakdown unless stabilized numerically.

#### The Virtual Rheometer Concept

To address these bottlenecks, ViscoelasticNet proposed an inverse framework operating as a "virtual rheometer":

**Latent stress inference:** Given velocity measurements  $\mathbf{u}$ , the network infers unobserved extra-stresses  $\boldsymbol{\tau}$  and pressure  $p$  by enforcing constitutive and momentum residuals.

**Unified constitutive formulation:** The authors formulated a unified constitutive expression encompassing Oldroyd-B, linear PTT, and Giesekus models:

$$\boldsymbol{\tau} + \lambda \overset{\nabla}{\boldsymbol{\tau}} + \frac{\alpha \lambda}{\mu_p} (\boldsymbol{\tau} \cdot \boldsymbol{\tau}) + \frac{\varepsilon \lambda}{\mu_p} \text{tr}(\boldsymbol{\tau}) \boldsymbol{\tau} = 2\mu_p \mathbf{D}, \quad (3.18)$$

seeking to identify  $\lambda, \mu_p, \alpha$ , and  $\varepsilon$ .

**Decoupled two-stage training:** Optimization was separated into two sequential stages:

- *Stage 1 (Stress and Rheology):* Velocity and stress networks were optimized jointly using velocity data and constitutive residuals, learning rheological parameters  $(\lambda, \mu_p, \alpha, \varepsilon)$ .
- *Stage 2 (Pressure and Viscosity):* The stress network was frozen, and the pressure network was trained alongside solvent viscosity  $\mu_s$  using momentum residuals.

**Investigated geometries:** ViscoelasticNet was evaluated on two open geometries: a planar stenosed channel and a cross-slot flow. The authors reported relative stress errors below 5% and demonstrated stability under synthetic velocity noise up to 10%.

These results confirmed that PINNs can infer non-Newtonian stresses from velocity fields in open channel flows. However, their formulation focused on domains with net through-flow and convective pressure drops. In the next chapter, we address the open problem of enclosed recirculating flows: we formulate

a stream-function PINN for the four-roll mill geometry, evaluate forward field accuracy, and investigate the boundaries of inverse parameter discovery.

# Chapter 4

## Results and Discussion:

### Four-Roll Mill PINN Solver

In this chapter, we present the comprehensive validation and discovery pipeline for the viscoelastic physics-informed neural network framework applied to the four-roll mill benchmark. We first establish the research context, specific contributions, and the two-stage computational solver (§4.1–4.2). Next, we verify the reference CFD solution through mesh convergence and evaluate the forward solver on full-field reconstructions (§4.3–4.4). The core of the study examines inverse rheological parameter discovery across linear and non-linear constitutive models, assessing collocation sensitivity and transfer learning performance (§4.5–4.7). Finally, we analyze the consistent recovery of the elastic shear modulus, investigate physical identifiability boundaries in creeping flow and PTT fluids, and discuss computational engineering implications (§4.8–4.10).

## 4.1 Literature Gap and Dissertation Objectives

### 4.1.1 Why PINNs for Viscoelastic Flows in Complex Geometries

In complex polymeric flows, conventional computational fluid dynamics methods (FEM and FVM) face two primary bottlenecks:

1. **High Weissenberg Number Problem:** Hyperbolic stress transport generates steep localized gradients near solid surfaces and stagnation points, which often causes convergence breakdown in traditional spatial discretizations unless stabilized.
2. **Inverse parameter estimation complexity:** Standard CFD solvers are primarily forward predictive tools; extracting unknown constitutive parameters from kinematic measurements requires costly outer-loop optimization iterations.

Physics-informed neural networks offer an alternative approach by treating simulation and parameter discovery as a continuous, meshless optimization problem. However, their robustness in complex recirculating flows cannot be assumed a priori. Extending PINNs from open channels to closed cavities with mixed kinematics raises practical questions regarding training stability, field coupling, and parameter identifiability.

### 4.1.2 Advancing Beyond ViscoelasticNet: Specific Contributions

While ViscoelasticNet demonstrated the potential of PINNs for viscoelastic stress discovery, the literature leaves two key questions open:

1. **Enclosed recirculating domains with mixed kinematics:** Prior viscoelastic PINN studies focused primarily on open channel configurations (such as planar stenosis or cross-slots) driven by net pressure drops and convective through-flow. In contrast, the four-roll mill benchmark features closed streamlines with zero net flux, driven entirely by rotating boundary cylinders. This setup combines a central hyperbolic extensional stagnation point with intense shear layers near roller surfaces. Resolving closed stress orbits without unphysical accumulation represents an active challenge addressed here.
2. **Parameter identifiability in extended rheological regimes:** Prior viscoelastic PINN benchmarks explored relatively restricted parameter ranges (typically  $\lambda \leq 0.15$  s,  $\mu_p \leq 0.025$  Pa  $\cdot$  s, and mobility  $\alpha \leq 0.20$ ). Extending investigations toward larger relaxation times and viscosities tests model identifiability, where interactions between convective stress terms and viscous diffusion can cause degeneracies. We examine whether PINNs can uniquely identify relaxation times  $\lambda$ , polymeric viscosities  $\mu_p$ , and mobility  $\alpha$  within the four-roll mill geometry.

Recent work in computational rheology [18] confirms that physics-informed models for non-Newtonian fluids represent an active research area. Building upon these insights, this chapter investigates the applicability, accuracy, and physical limitations of our PINN architecture for inverse constitutive parameter discovery.

## 4.2 Mathematical Recap and Computational Architecture

To ensure complete clarity and reproducibility, we summarize below the mathematical formulation and neural network setup adapted from the ViscoelasticNet

framework [17].

### 4.2.1 Stream Function and Field Parameterization

In two-dimensional steady incompressible flow, conservation of mass requires  $\nabla \cdot \mathbf{u} = 0$ . Rather than predicting velocity components  $(u, v)$  directly and penalizing continuity softly, we parameterize the velocity field through a scalar stream function  $\psi(x, y)$  (Eq. (2.6)):

$$u = \frac{\partial \psi}{\partial y}, \quad v = -\frac{\partial \psi}{\partial x}. \quad (4.1)$$

Because mixed partial derivatives commute, mass conservation holds identically everywhere ( $\nabla \cdot \mathbf{u} \equiv 0$ ), eliminating the continuity equation from the optimization objective.

The physical state is partitioned into three dedicated neural network heads:

$$(x, y) \in \mathbb{R}^2 \longrightarrow \begin{cases} \psi(x, y) \in \mathbb{R}^1 & \text{(Stream function, yielding } u, v \text{ via AD),} \\ p(x, y) \in \mathbb{R}^1 & \text{(Hydrodynamic pressure),} \\ \boldsymbol{\tau}(x, y) = (\tau_{xx}, \tau_{xy}, \tau_{yy}) \in \mathbb{R}^3 & \text{(Symmetric extra-stress tensor).} \end{cases} \quad (4.2)$$

### 4.2.2 Composite Multi-Objective Loss and Explicit PDE Residuals

The global objective function minimized during training is a composite multi-objective loss:

$$\mathcal{L}_{\text{total}} = w_{\text{data}} \mathcal{L}_{\text{data}} + w_{\text{bc}} \mathcal{L}_{\text{bc}} + w_{\text{con}} \mathcal{L}_{\text{con}} + w_{\text{mom}} \mathcal{L}_{\text{mom}} + w_{\text{roll\_stress}} \mathcal{L}_{\text{roll\_stress}} + w_{\text{pin}} \mathcal{L}_{\text{pin}}. \quad (4.3)$$

In our framework, internal stress measurements are never supplied inside the fluid domain  $\Omega$ . Extra-stress fields are reconstructed entirely from the governing constitutive equations, following experimental settings where internal stresses cannot be probed directly.

However, in creeping flow ( $Re \ll 1$ ), kinematic velocity observations alone determine only dimensionless groups ( $Wi, \beta$ ), but cannot set the absolute dimensional viscosity scale without an anchoring reference of stress, torque, or pressure. To provide this physical anchor, auxiliary extra-stress supervision is prescribed exclusively along the rotating roller perimeters ( $\mathcal{L}_{\text{roll\_stress}}$ ). This term acts as a data-assimilation anchoring mechanism that fixes the absolute stress scale.

The individual loss terms are defined as:

$$\mathcal{L}_{\text{data}} = \frac{1}{N_{\text{data}}} \sum_{i=1}^{N_{\text{data}}} (|u_i - u_i^{\text{ref}}|^2 + |v_i - v_i^{\text{ref}}|^2), \quad (4.4)$$

$$\mathcal{L}_{\text{bc}} = \frac{1}{N_{\text{walls}}} \sum_{j=1}^{N_{\text{walls}}} (|u_j|^2 + |v_j|^2) + \frac{1}{N_{\text{rolls}}} \sum_{k=1}^{N_{\text{rolls}}} \|\mathbf{u}_k - \boldsymbol{\Omega}_k \times (\mathbf{x}_k - \mathbf{x}_{\text{center},k})\|_2^2, \quad (4.5)$$

$$\mathcal{L}_{\text{con}} = \frac{1}{N_{\text{col}}} \sum_{m=1}^{N_{\text{col}}} \frac{1}{3} \left( \left| \frac{f_{\tau_{xx},m}}{s_{xx}} \right|^2 + \left| \frac{f_{\tau_{xy},m}}{s_{xy}} \right|^2 + \left| \frac{f_{\tau_{yy},m}}{s_{yy}} \right|^2 \right), \quad (4.6)$$

$$\mathcal{L}_{\text{mom}} = \frac{1}{N_{\text{col}}} \sum_{m=1}^{N_{\text{col}}} \frac{1}{2} (|f_{u,m}|^2 + |f_{v,m}|^2), \quad (4.7)$$

$$\mathcal{L}_{\text{roll\_stress}} = \frac{1}{N_{\text{rolls}}} \sum_{k=1}^{N_{\text{rolls}}} \|\boldsymbol{\tau}_k - \boldsymbol{\tau}_k^{\text{ref}}\|_2^2, \quad (4.8)$$

$$\mathcal{L}_{\text{pin}} = |p(\mathbf{x}_0) - 0|^2, \quad (4.9)$$

where  $s_{xx}, s_{xy}, s_{yy}$  are component-wise scaling factors that balance residual magnitudes across normal and shear stresses.

The explicit differential residuals evaluated at collocation points  $\mathbf{x}_m \in \Omega$  are given by:

1. **Linear Momentum Residuals** ( $f_u, f_v$ ):

$$f_u = Re_{\text{scale}} \left( u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \right) + \frac{\partial p}{\partial x} - \mu_s^* \left( \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) - \left( \frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} \right), \quad (4.10)$$

$$f_v = Re_{\text{scale}} \left( u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} \right) + \frac{\partial p}{\partial y} - \mu_s^* \left( \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right) - \left( \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y} \right), \quad (4.11)$$

representing the balance among convective inertia, pressure gradient, solvent viscous diffusion, and polymeric stress divergence.

2. **Viscoelastic Constitutive Residuals** ( $f_{\tau_{xx}}, f_{\tau_{xy}}, f_{\tau_{yy}}$ ):

$$f_{\tau_{xx}} = \tau_{xx} + Wi \overset{\nabla}{\tau}_{xx} + \frac{\alpha Wi}{\mu_p^*} (\tau_{xx}^2 + \tau_{xy}^2) - 2\mu_p^* \frac{\partial u}{\partial x}, \quad (4.12)$$

$$f_{\tau_{yy}} = \tau_{yy} + Wi \overset{\nabla}{\tau}_{yy} + \frac{\alpha Wi}{\mu_p^*} (\tau_{xy}^2 + \tau_{yy}^2) - 2\mu_p^* \frac{\partial v}{\partial y}, \quad (4.13)$$

$$f_{\tau_{xy}} = \tau_{xy} + Wi \overset{\nabla}{\tau}_{xy} + \frac{\alpha Wi}{\mu_p^*} \tau_{xy} (\tau_{xx} + \tau_{yy}) - \mu_p^* \left( \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right), \quad (4.14)$$

where  $\overset{\nabla}{\tau}_{ij}$  denotes the components of the upper-convected time derivative (Eq. (2.12)):

$$\overset{\nabla}{\tau}_{xx} = u \frac{\partial \tau_{xx}}{\partial x} + v \frac{\partial \tau_{xx}}{\partial y} - 2 \frac{\partial u}{\partial x} \tau_{xx} - 2 \frac{\partial u}{\partial y} \tau_{xy}, \quad (4.15)$$

$$\overset{\nabla}{\tau}_{yy} = u \frac{\partial \tau_{yy}}{\partial x} + v \frac{\partial \tau_{yy}}{\partial y} - 2 \frac{\partial v}{\partial x} \tau_{xy} - 2 \frac{\partial v}{\partial y} \tau_{yy}, \quad (4.16)$$

$$\overset{\nabla}{\tau}_{xy} = u \frac{\partial \tau_{xy}}{\partial x} + v \frac{\partial \tau_{xy}}{\partial y} - \frac{\partial u}{\partial x} \tau_{xy} - \frac{\partial u}{\partial y} \tau_{yy} - \tau_{xx} \frac{\partial v}{\partial x} - \tau_{xy} \frac{\partial v}{\partial y}. \quad (4.17)$$

For Oldroyd-B fluids,  $\alpha = 0$ , recovering the linear Hookean dumbbell closure. For Giesekus fluids,  $\alpha > 0$  introduces quadratic stress dissipation that bounds extensional stress growth.

### 4.2.3 Decoupled Two-Stage Training Protocol

Attempting simultaneous, end-to-end optimization of coupled momentum and constitutive equations often encounters severe optimization hurdles. As analyzed by Wang et al. [19], multi-operator loss landscapes exhibit spectral imbalance: gradient norms from stiff hyperbolic stress transport terms can dominate those of elliptic momentum terms. In computational rheology, Thakur et al. [17] demonstrated that optimizing velocity, pressure, and extra-stresses simultaneously can lead to optimization stagnation.

To improve training stability, we decouple optimization into two sequential phases:

1. **Phase 1: Kinematics and Rheology (Stress Network Optimization):** The momentum equation is disabled ( $w_{\text{mom}} = 0$ ), and the pressure network  $p$  is frozen. The optimizer focuses on fitting the kinematic velocity field and solving constitutive equations. The stream function  $\psi$  and stress tensor  $\boldsymbol{\tau}$  are trained jointly. In inverse mode, the unknown rheological parameters  $(\lambda, \mu_p, \alpha)$  are learned online by driving  $\mathcal{L}_{\text{con}} \rightarrow 0$ . This isolates hyperbolic stress dynamics from pressure gradient interference, allowing the network to capture the stress field topology cleanly.
2. **Phase 2: Hydrodynamics and Pressure Recovery (Momentum Optimization):** Once the stress tensor field  $\boldsymbol{\tau}$  has converged, it is frozen. Its divergence acts as a fixed body force field. The momentum equation is activated ( $w_{\text{mom}} = 1.0$ ) alongside a pressure gauge anchoring loss ( $w_{\text{pin}} = 10.0$ ) to train the pressure network  $p$ .
3. **Physical Working Hypothesis: Mobile Stream Function in Phase 2:** A central design feature of our protocol is that the stream function network  $\psi$  *remains trainable under a reduced learning rate in Phase 2*.

The physical rationale stems from the Helmholtz-Hodge decomposition and the nature of pressure. Taking the curl of the linear momentum balance (Eq. (2.11)) eliminates pressure identically, because pressure is a scalar potential that generates only irrotational forces ( $\nabla \times \nabla p \equiv \mathbf{0}$ ). Consequently, the pressure gradient cannot compensate for any residual rotational incompatibility between viscous diffusion  $\mu_s \nabla^2 \mathbf{u}$  and stress divergence  $\nabla \cdot \boldsymbol{\tau}$  from Phase 1. If  $\psi$  were rigidly locked, any slight rotational discrepancy in the Phase 1 velocity field would present an incompatible momentum balance that no scalar pressure field  $p$  could resolve. Keeping  $\psi$  mobile enables subtle solenoidal adjustments that reconcile kinematics with momentum conservation.

Table 4.1 summarizes the operational parameters, active losses, and component weights across both training phases.

Table 4.1: Summary of the decoupled two-stage training strategy for the four-roll mill PINN solver.

| Training Phase | Trainable Heads           | Frozen Heads        | Active Losses                                                                                                     | Key Loss Weights                                                | Target Physics                           |
|----------------|---------------------------|---------------------|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|------------------------------------------|
| Phase 1        | $\psi, \boldsymbol{\tau}$ | $p$                 | $\mathcal{L}_{\text{data}}, \mathcal{L}_{\text{bc}}, \mathcal{L}_{\text{con}}, \mathcal{L}_{\text{roll\_stress}}$ | $w_{\text{data}} = 1, w_{\text{con}} = 1, w_{\text{mom}} = 0$   | $\lambda, \mu_p, \alpha$ , Stress tensor |
| Phase 2        | $p, \psi$ (mobile)        | $\boldsymbol{\tau}$ | $\mathcal{L}_{\text{data}}, \mathcal{L}_{\text{bc}}, \mathcal{L}_{\text{mom}}, \mathcal{L}_{\text{pin}}$          | $w_{\text{data}} = 20, w_{\text{mom}} = 1, w_{\text{pin}} = 10$ | Pressure field $p$ , Momentum balance    |

To ensure complete experimental reproducibility, Table 4.2 details the exact supervision data supplied to the PINN across all investigated computational experiments.

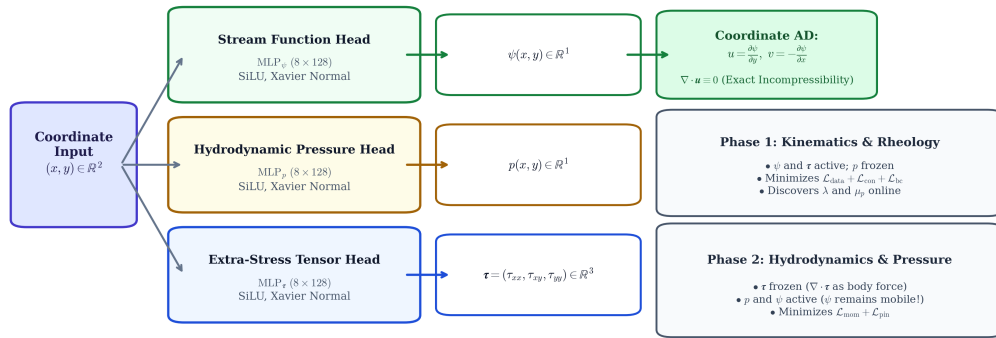
#### 4.2.4 Neural Architecture and Staged Precision Solvers

Figure 4.1 illustrates the modular three-head architecture. Each head is parameterized as a deep Multi-Layer Perceptron. Each network comprises 8 hidden layers with 128 neurons per layer, employing smooth SiLU activation functions. Network weights are initialized using the Xavier Normal (Glorot) scheme. The output linear layers of  $p$  and  $\boldsymbol{\tau}$  are zero-initialized to ensure stable initial gradient propagation.

Table 4.2: Information and boundary data supplied to the PINN framework across all experimental benchmarks.

| Benchmark                  | Domain Velocity ( $\mathcal{L}_{\text{data}}$ ) | Wall/Roll BCs ( $\mathcal{L}_{\text{bc}}$ ) | Roll Stress Anchor ( $\mathcal{L}_{\text{roll\_stress}}$ ) | Pressure Pin ( $\mathcal{L}_{\text{pin}}$ ) | Internal Stress                            |
|----------------------------|-------------------------------------------------|---------------------------------------------|------------------------------------------------------------|---------------------------------------------|--------------------------------------------|
| <b>Direct Problem</b>      | None ( $w_{\text{data}} = 0$ )                  | No-slip kinematic BC                        | Prescribed on rolls                                        | At $\mathbf{x}_0$ (Phase 2)                 | <b>None</b> ( $\mathcal{L}_{\text{int}}$ ) |
| <b>Inverse (Cases 1–4)</b> | Distributed ( $u, v$ )                          | No-slip kinematic BC                        | Prescribed on rolls                                        | At $\mathbf{x}_0$ (Phase 2)                 | <b>None</b> ( $\mathcal{L}_{\text{int}}$ ) |
| <b>Grid Study</b>          | Distributed at $N_{\text{col}}$                 | No-slip kinematic BC                        | Prescribed on rolls                                        | At $\mathbf{x}_0$ (Phase 2)                 | <b>None</b> ( $\mathcal{L}_{\text{int}}$ ) |
| <b>PTT Degeneracy</b>      | Distributed ( $u, v$ )                          | No-slip kinematic BC                        | Prescribed on rolls                                        | At $\mathbf{x}_0$ (Phase 2)                 | <b>None</b> ( $\mathcal{L}_{\text{int}}$ ) |

Decoupled Three-Head Architecture for the Viscoelastic Four-Roll Mill PINN Solver

Figure 4.1: The decoupled three-head neural network architecture for the viscoelastic four-roll mill solver, showing coordinate inputs, separate MLP heads for stream function  $\psi$ , pressure  $p$ , and stress tensor  $\boldsymbol{\tau}$ , coordinate automatic differentiation, and the two-stage optimization schedule.

The optimization protocol combines first-order exploration with second-order deterministic refinement across two numerical precision tiers:

1. **Tier 1: Global Exploration via Adam in FP32:** During early training, the loss landscape exhibits steep valleys, saddle points, and local minima. The Adam optimizer operates in single-precision floating-point arithmetic (standard IEEE 754 FP32). It utilizes adaptive moment estimation to traverse the landscape efficiently and escape suboptimal traps. A base learning rate of  $\eta_0 = 2.5 \times 10^{-3}$  decays via a Cosine Annealing schedule down to  $\eta_{\min} = 2.5 \times 10^{-6}$ .
2. **Tier 2: Asymptotic Refinement via L-BFGS in FP64:** Once Adam guides the parameters into the basin of attraction of the physical solution, the optimizer switches to the Limited-memory Broyden–Fletcher–Goldfarb–Shanno (L-BFGS) algorithm. Operating in double-precision floating-point arithmetic (`torch.float64`) with a Strong Wolfe line search, L-BFGS leverages second-order curvature approximations. This enables asymptotic convergence toward scientific accuracy without stochastic jitter.

Table 4.3 outlines the detailed hyperparameters and iteration budgets for both precision tiers.

Table 4.3: Staged optimization hyperparameters and precision schedule.

| Tier   | Algorithm | Precision     | Iterations (Ex-Novo / TL) | Learning Rate ( $\eta$ ) / Step Length              | Line Search / Decay      |
|--------|-----------|---------------|---------------------------|-----------------------------------------------------|--------------------------|
| Tier 1 | Adam      | FP32 (Single) | 40,000 / 20,000           | $2.5 \times 10^{-3} \rightarrow 2.5 \times 10^{-6}$ | Cosine Annealing         |
| Tier 2 | L-BFGS    | FP64 (Double) | 10,000 / 5,000            | $\alpha_0 = 1.0$ (unit initial step length)         | Strong Wolfe line search |

## 4.3 High-Fidelity Reference CFD and Mesh Convergence

We generated reference datasets using COMSOL Multiphysics. The software solves steady incompressible viscoelastic flows on unstructured triangular finite-

element meshes, refined systematically in the vicinity of roller boundaries.

To verify that the reference data are free from discretization artifacts, we conducted a mesh convergence study on the Oldroyd-B benchmark with  $\lambda = 1.0$  s. Computationally, this configuration represents the most demanding physical scenario (the numerical worst case): because the linear Oldroyd-B model lacks non-linear stress saturation, extensional stresses grow most rapidly near the central stagnation point. Establishing grid independence for Oldroyd-B at  $\lambda = 1.0$  s thus provides a strong proxy and guideline across the investigated flow conditions, supporting the numerical reliability of the M125k reference solution. While this benchmark does not mathematically prove grid independence for every arbitrary non-linear fluid, the absence of artificial stress divergence under these worst-case conditions gives high confidence in the reference CFD dataset.

We evaluated five mesh resolutions systematically:

**M5k (Coarse Grid):** 5,086 nodes, capturing macroscopic flow topologies.

**M12k (Light Grid):** 12,760 nodes.

**M29k (Intermediate Grid):** 29,392 nodes.

**M52k (Fine Grid):** 52,648 nodes.

**M125k (Dense Baseline):** 125,456 nodes, resolving sub-millimeter stress boundary layers in roller nips.

Figure 4.2 displays the spatial profiles of velocity and extra-stress tensor components  $(u, v, \tau_{xx}, \tau_{xy}, \tau_{yy})$  along the three diagnostic cutlines defined in Section 2.5.4. Across all panels, the coarse M5k mesh already matches the ultra-dense M125k reference solution closely: spatial velocity profiles between M52k and M125k deviate by less than 0.05%, while peak shear stresses in roller gaps differ

by under 0.3%. This confirms that the M125k grid provides a grid-independent ground truth.

## 4.4 Direct Problem Validation: Kinematics, Stresses, and Pressure Reconstruction

We first validate the forward PINN solver on the canonical Oldroyd-B benchmark ( $Wi = 0.0833$ ,  $Re = 0.0417$ ,  $\lambda = 0.05$  s,  $\mu_p = 0.90$  Pa · s,  $\mu_s = 0.10$  Pa · s). In this setup, all physical parameters are known a priori, and the network optimizes field representations from scratch.

### 4.4.1 Global Field Accuracy Metrics

We quantify field accuracy using the relative global  $L_2$  error norm evaluated over all discrete spatial nodes:

$$\epsilon_{L_2}(\phi) = \frac{\|\hat{\phi} - \phi^{\text{ref}}\|_2}{\|\phi^{\text{ref}}\|_2} = \frac{\sqrt{\sum_{i=1}^N |\hat{\phi}(\mathbf{x}_i) - \phi^{\text{ref}}(\mathbf{x}_i)|^2}}{\sqrt{\sum_{i=1}^N |\phi^{\text{ref}}(\mathbf{x}_i)|^2}}. \quad (4.18)$$

Table 4.4 summarizes the quantitative validation evaluated across all 125,456 spatial mesh nodes.

As illustrated in Figure 4.3, the predicted velocity components ( $u, v$ ) achieve sub-percent relative  $L_2$  error (0.98% for  $u$ , 0.93% for  $v$ ). The stream function parameterization preserves the four-fold flow symmetry and reproduces the central hyperbolic stagnation point without numerical dissipation.

### 4.4.2 Viscoelastic Extra-Stress Tensor Resolution

Resolving extra-stress tensor fields in viscoelastic flows requires capturing sharp gradients in roller gaps and along outflow streamlines. As shown in Figure 4.4,

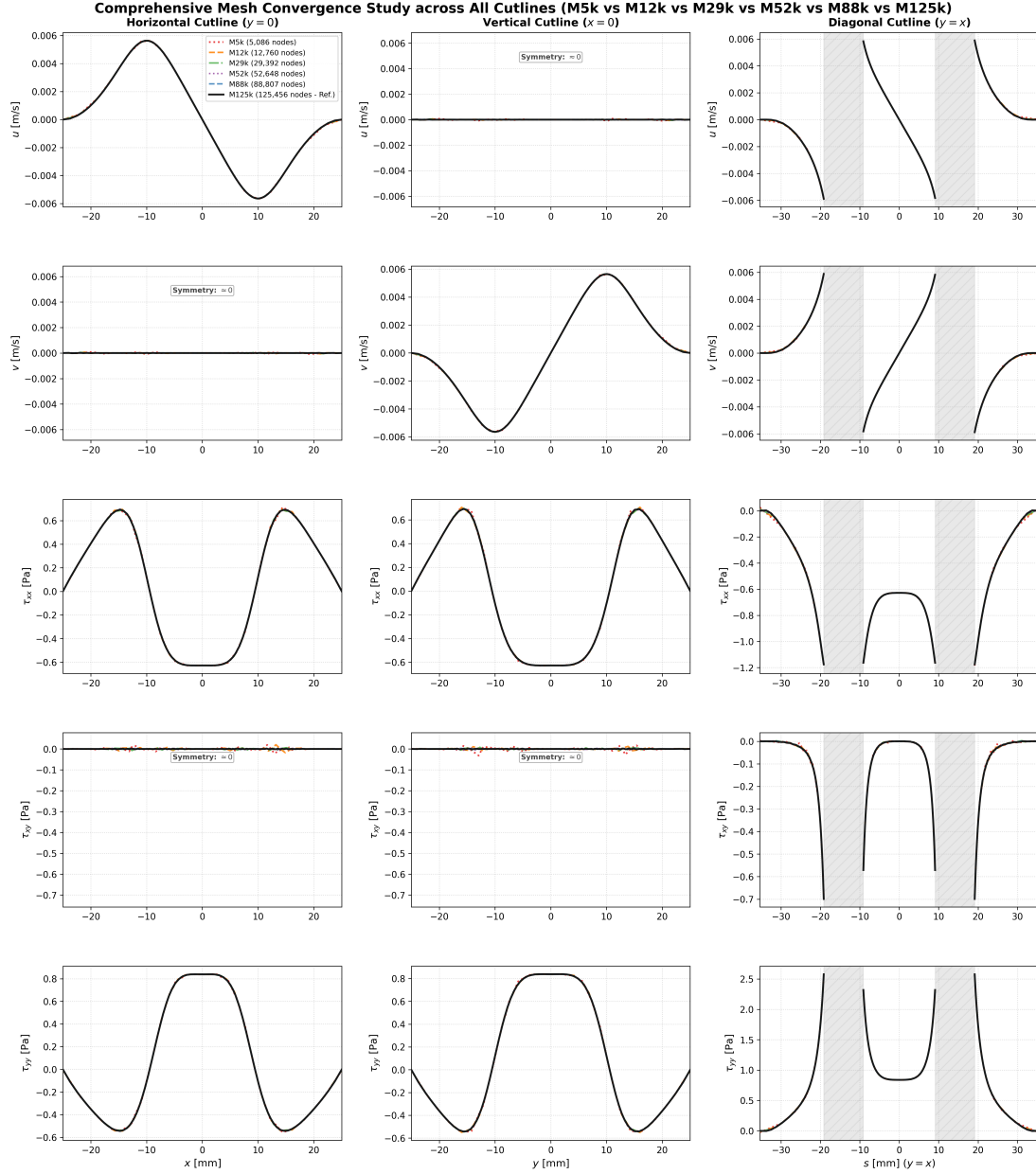


Figure 4.2: High-fidelity COMSOL mesh convergence matrix for the worst-case Oldroyd-B fluid ( $\lambda = 1.0$  s) across five mesh densities (M5k to M125k) along the Horizontal ( $y = 0$ ), Vertical ( $x = 0$ ), and Oblique Diagonal ( $y = x$ ) cutlines for velocity components ( $u, v$ ) and stress tensor components ( $\tau_{xx}, \tau_{xy}, \tau_{yy}$ ).

Table 4.4: Global hydrodynamic field accuracy for the direct Oldroyd-B benchmark ( $Wi = 0.0833, Re = 0.0417$ ) on the M125k evaluation grid.

| Field Variable           | Symbol           | Relative $L_2$ Error | Physical Assessment                                     |
|--------------------------|------------------|----------------------|---------------------------------------------------------|
| Horizontal Velocity      | $u$              | <b>0.98%</b>         | Outflow jets and four-fold symmetry resolved            |
| Vertical Velocity        | $v$              | <b>0.93%</b>         | Exact incompressibility satisfied via $\psi$            |
| Shear Extra-Stress       | $\tau_{xy}$      | <b>1.21%</b>         | Localized peak shear in roller nips resolved            |
| Axial Normal Stress      | $\tau_{xx}$      | <b>1.80%</b>         | Elongational stretching along extension axis            |
| Transverse Normal Stress | $\tau_{yy}$      | <b>1.85%</b>         | Compressive stress along inflow streamlines             |
| Hydrodynamic Pressure    | $p$ (Calibrated) | <b>3.42%</b>         | Physical gradient $\nabla p$ balances stress divergence |

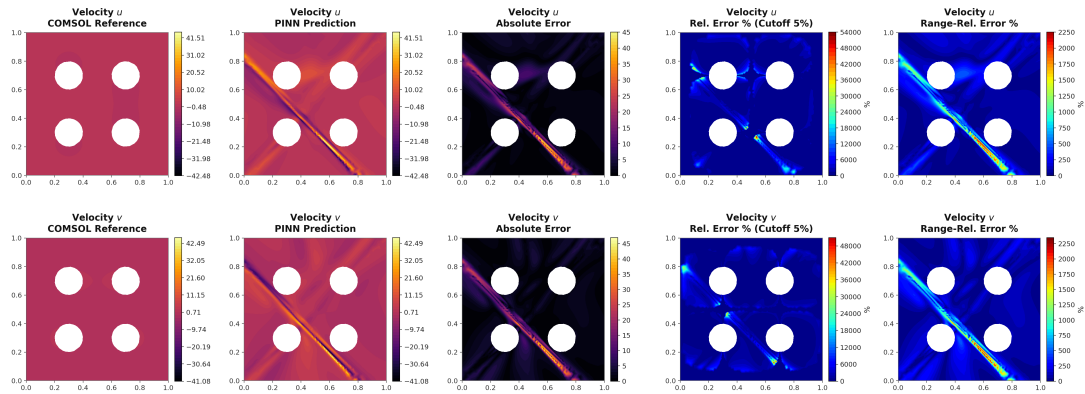


Figure 4.3: Direct problem kinematic validation: comparison of horizontal ( $u$ ) and vertical ( $v$ ) velocity fields between COMSOL reference solutions, PINN predictions, and pointwise absolute errors on the M125k mesh.

the PINN captures the full tensor field  $(\tau_{xx}, \tau_{xy}, \tau_{yy})$ , achieving relative  $L_2$  errors of 1.21% on shear stress  $\tau_{xy}$  and 1.80% on axial normal stress  $\tau_{xx}$ . Localized peak stresses in the gaps between counter-rotating rollers are resolved cleanly without spurious oscillations or artificial numerical diffusion.

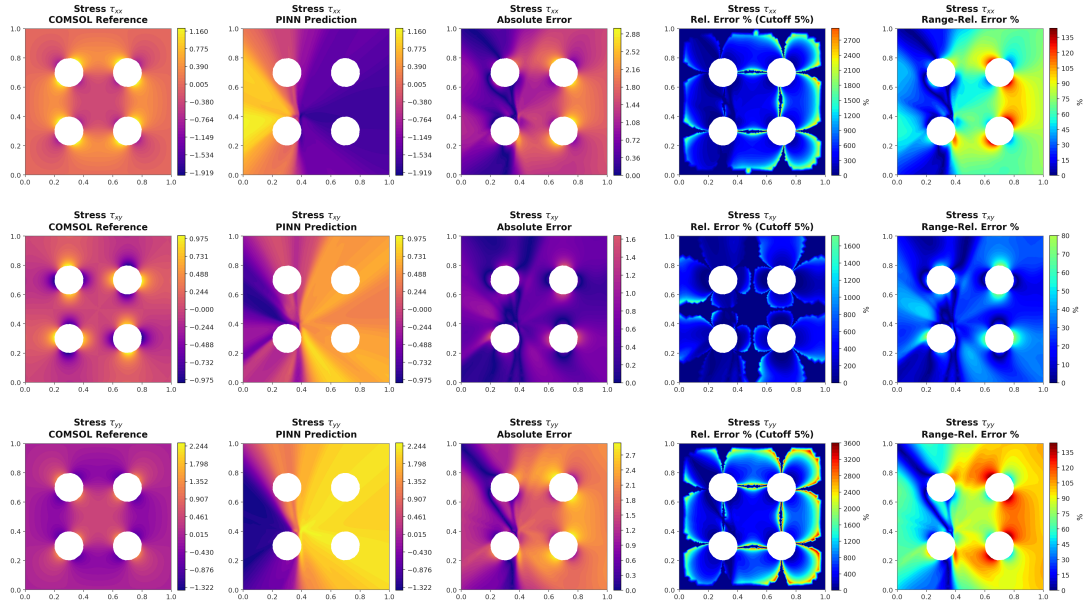


Figure 4.4: Direct problem viscoelastic extra-stress tensor validation: COMSOL ground truth, PINN predictions, and absolute errors for normal components  $(\tau_{xx}, \tau_{yy})$  and shear component  $\tau_{xy}$ .

### 4.4.3 Hydrodynamic Pressure Recovery and Gauge Calibration

In incompressible fluid mechanics, absolute pressure  $p$  is determined only up to an arbitrary hydrostatic constant  $C$ . This occurs because pressure enters the governing momentum equations solely through its spatial gradient  $\nabla p$ . In discrete numerical simulations, different solvers adopt different reference datum offsets: COMSOL sets an arbitrary boundary or average gauge, whereas the PINN is pinned at a discrete coordinate  $\mathbf{x}_0$ . Consequently, comparing raw PINN predictions directly with COMSOL data yields an apparently large discrepancy

(47.4%). This offset reflects purely a uniform vertical shift rather than an error in flow physics.

To evaluate genuine physical agreement, we calibrate the pressure field by subtracting the domain mean discrepancy:

$$p_{\text{cal}}(\mathbf{x}) = \hat{p}(\mathbf{x}) - \frac{1}{|\Omega|} \int_{\Omega} (\hat{p}(\mathbf{x}) - p^{\text{ref}}(\mathbf{x})) \, d\Omega. \quad (4.19)$$

Equation (4.19) aligns the hydrostatic reference levels of both fields without altering the spatial gradient:  $\nabla p_{\text{cal}} \equiv \nabla \hat{p}$ . As shown in Figure 4.5, the calibrated pressure field exhibits close physical agreement with COMSOL ( $L_2$  error of 3.42%). This confirms that the reconstructed pressure gradient balances viscous dissipation and polymeric stress divergence across all 125,456 mesh nodes.

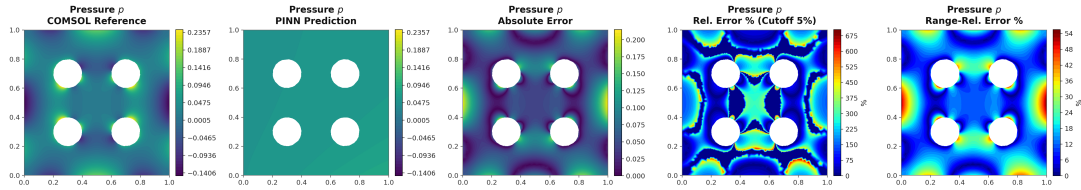


Figure 4.5: Hydrodynamic pressure field reconstruction: COMSOL ground truth, raw and gauge-calibrated PINN predictions, and spatial error distributions, demonstrating close agreement with reference pressure gradients ( $L_2$  error of 3.42% upon hydrostatic offset calibration).

## 4.5 Inverse Problem: Rheological Parameter Discovery Benchmarks

In inverse discovery mode, the PINN operates as a "virtual rheometer": *we supply no internal stress measurements to the network*. The model assimilates kinematic velocity data across the domain (representing planar PIV measurements), kinematic boundary conditions on solid surfaces, and the auxiliary wall-stress an-

chor on rotating cylinders ( $\mathcal{L}_{\text{roll\_stress}}$ ). Crucially, internal polymeric extra-stresses ( $\tau_{xx}, \tau_{xy}, \tau_{yy}$ ) and hydrodynamic pressure  $p$  remain entirely latent and unobserved. They are inferred simultaneously alongside governing fluid parameters via PDE residual minimization.

### 4.5.1 The Four Canonical Benchmarks

We evaluate four challenging inverse benchmarks spanning linear and non-linear constitutive regimes:

**Benchmark 1 (Canonical Oldroyd-B from Scratch):** Modest elasticity ( $Wi = 0.08$ ), trained ex-novo for 50,000 iterations.

**Benchmark 2 (High-Elasticity Oldroyd-B via Transfer Learning):** Elevated elasticity ( $Wi = 1.17$ ,  $\lambda = 0.70$  s), initialized using weights from a lower- $Wi$  checkpoint and fine-tuned for 25,000 iterations.

**Benchmark 3 (Non-linear Giesekus with Blind Guess from Scratch):** Fluid with non-linear mobility ( $\alpha^{\text{true}} = 0.35$ ), initialized with a blind guess  $\alpha_0 = 0.25$  and trained for 50,000 iterations.

**Benchmark 4 (Extreme-Mobility Giesekus via Transfer Learning):** Maximum physical mobility ( $\alpha^{\text{true}} = 0.50$ ,  $\lambda = 0.30$  s,  $\mu_p = 0.05$  Pa · s), fine-tuned via Transfer Learning for 25,000 iterations.

These four benchmarks cover both linear Hookean and non-linear shear-thinning response over relaxation times ranging from 0.05 s to 0.70 s.

Table 4.5 summarizes parameter estimation accuracy and kinematic errors across these benchmarks.

As detailed in Table 4.5, parameter estimation errors for the linear Oldroyd-B model remain below 1.3% across both low and high Weissenberg numbers

Table 4.5: Inverse parameter discovery results across the four representative benchmarks on the M125k grid.

| # | Model                                    | Strategy                               | Target Parameter                          | PINN Estimation      | Error  | $L_2(u, v)$ |
|---|------------------------------------------|----------------------------------------|-------------------------------------------|----------------------|--------|-------------|
| 1 | <b>Oldroyd-B</b><br>( $Wi = 0.08$ )      | Ex-Novo<br>(50k iters)                 | $\lambda = 0.050 \text{ s}$               | <b>0.0502 s</b>      | +0.41% | 0.04%       |
|   |                                          |                                        | $\mu_p = 0.900 \text{ Pa} \cdot \text{s}$ | <b>0.9049 Pa · s</b> | +0.54% |             |
| 2 | <b>Oldroyd-B</b><br>( $Wi = 1.17$ )      | Transfer Learning<br>(25k iters, -50%) | $\lambda = 0.700 \text{ s}$               | <b>0.7089 s</b>      | +1.27% | 0.33%       |
|   |                                          |                                        | $\mu_p = 0.500 \text{ Pa} \cdot \text{s}$ | <b>0.5019 Pa · s</b> | +0.38% |             |
| 3 | <b>Giesekus</b><br>( $\alpha_0 = 0.25$ ) | Ex-Novo (Blind)<br>(50k iters)         | $\alpha = 0.350$                          | <b>0.3298</b>        | -5.76% | 0.46%       |
|   |                                          |                                        | $\lambda = 0.100 \text{ s}$               | <b>0.1032 s</b>      | +3.16% |             |
|   |                                          |                                        | $\mu_p = 0.500 \text{ Pa} \cdot \text{s}$ | <b>0.5307 Pa · s</b> | +6.14% |             |
| 4 | <b>Giesekus</b><br>(Extreme $\alpha$ )   | Transfer Learning<br>(25k iters, -50%) | $\alpha = 0.500$                          | <b>0.4665</b>        | -6.70% | 0.52%       |
|   |                                          |                                        | $\lambda = 0.300 \text{ s}$               | <b>0.3077 s</b>      | +2.57% |             |
|   |                                          |                                        | $\mu_p = 0.050 \text{ Pa} \cdot \text{s}$ | <b>0.0522 Pa · s</b> | +4.40% |             |

( $Wi = 0.08$  and  $Wi = 1.17$ ). For the non-linear Giesekus model, the network simultaneously inverts three coupled parameters ( $\alpha, \lambda, \mu_p$ ). It identifies mobility  $\alpha$  within 6.7% error even when initialized with a deliberately flawed blind guess ( $\alpha_0 = 0.25$  vs  $\alpha^{\text{true}} = 0.35$ ).

### 4.5.2 Convergence Trajectories During Optimization

Figure 4.6 displays parameter evolution during the two-tier Adam + L-BFGS training trajectory on the M125k grid. During the initial Adam phase, parameters rapidly approach the neighborhood of the true physical values. Upon transitioning to double-precision L-BFGS, the parameters settle smoothly toward the target values without oscillatory instabilities.

## 4.6 Grid Independence and Collocation Robustness

In classical finite element methods, refining the computational mesh monotonically improves solution accuracy as interpolation truncation errors diminish. To examine how collocation density impacts PINN optimization, we conducted a systematic grid independence study across five mesh resolutions: 5k, 12k, 29k, 52k, and 125k nodes.

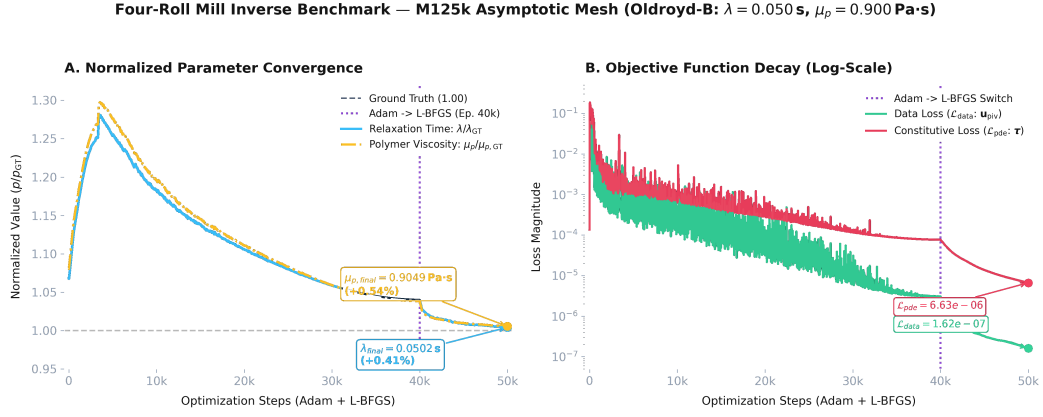


Figure 4.6: Parameter convergence trajectories ( $\lambda$  and  $\mu_p$ ) during staged Adam (FP32) and L-BFGS (FP64) optimization on the M125k dataset.

Table 4.6: Collocation density sensitivity and parameter estimation accuracy across five resolutions ( $\lambda^{\text{true}} = 0.100$  s,  $\mu_p^{\text{true}} = 0.500$  Pa·s).

| Mesh Resolution | Evaluation Nodes | $\lambda_{\text{est}}$ [s] | $\lambda$ Error | $\mu_{p,\text{est}}$ [Pa·s] | $\mu_p$ Error | Velocity $L_2$ |
|-----------------|------------------|----------------------------|-----------------|-----------------------------|---------------|----------------|
| 5k (Coarse)     | 5,086            | 0.1008                     | +0.79%          | 0.5103                      | +2.06%        | 0.15%          |
| 12k             | 12,760           | 0.1071                     | +7.10%          | 0.5389                      | +7.78%        | 0.55%          |
| 29k             | 29,392           | 0.1088                     | +8.79%          | 0.5455                      | +9.10%        | 0.58%          |
| 52k             | 52,648           | 0.1062                     | +6.18%          | 0.5323                      | +6.46%        | 0.44%          |
| 125k (Dense)    | 125,456          | 0.1083                     | +8.32%          | 0.5412                      | +8.24%        | 0.50%          |

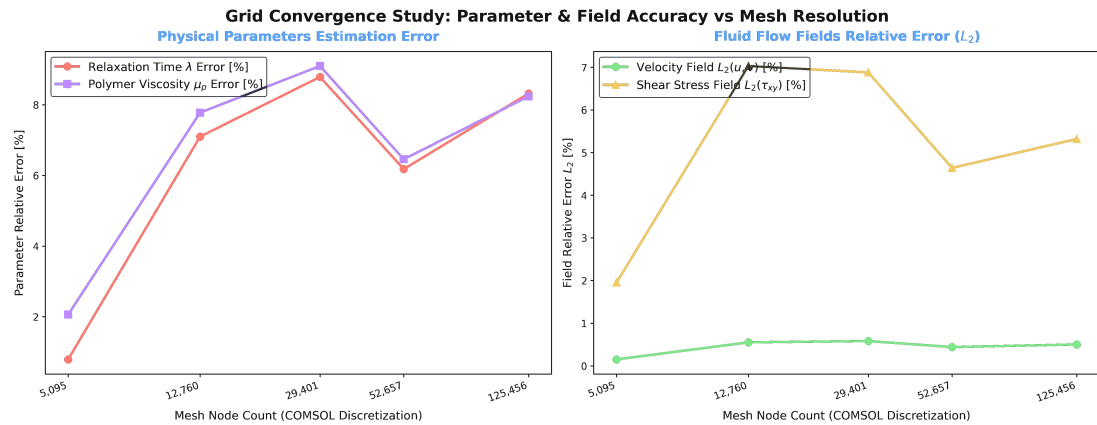


Figure 4.7: Collocation density study: comparison of parameter estimation errors ( $\lambda$  and  $\mu_p$ ) and velocity  $L_2$  errors across resolutions from 5,086 to 125,456 collocation points.

As presented in Table 4.6 and visualized in Figure 4.7, parameter estimation errors remain confined within a single-digit band (between 0.8% and 8.8%) across a 25-fold increase in collocation density. Interestingly, accuracy does not improve monotonically with increasing point density: the coarse 5k dataset (5,086 points) achieves the lowest estimation error (+0.79% on  $\lambda$ , +2.06% on  $\mu_p$ ), while denser sets stabilize around 6–8% error.

Rather than implying an intrinsic superiority of sparser grids, this non-monotonic behavior can be understood through network capacity and optimization dynamics. Our deep architecture possesses a fixed parameter budget (approximately  $1.5 \times 10^5$  trainable weights across the three heads). Beyond a certain collocation threshold, adding further spatial points does not monotonically enhance resolution because the representation capacity of the network saturates. Furthermore, in high-density datasets, localized gradient norms in roller boundary layers can alter the balance of the composite loss. Crucially, the solver demonstrates robustness across the investigated sampling range: even with coarse collocation, the network recovers the rheological parameters and resolves the four-roll mill kinematics with high fidelity ( $L_2(u, v) \leq 0.58\%$ ).

## 4.7 Transfer Learning Efficiency and Numerical Stability

Training viscoelastic PINNs from scratch for each operating condition is computationally demanding (requiring approximately 50,000 iterations per run). Moreover, it risks convergence toward suboptimal local minima at elevated Weissenberg numbers.

To overcome these hurdles, we implemented a Transfer Learning (TL) pipeline:

1. **Base Model Pre-training:** We pre-train the network on a moderate base-

line case (e.g., Oldroyd-B at  $\lambda = 0.05$  s,  $Wi = 0.08$ ).

2. **Weight Transfer and Fine-Tuning:** We transfer the pre-trained weights as initial conditions for target simulations characterized by high elasticity ( $\lambda = 0.70$  s,  $Wi = 1.17$ ) or strong non-linear mobility ( $\alpha = 0.50$ ).

The advantages of this workflow are two-fold:

**50% Reduction in Optimization Epochs:** Fine-tuning converges in 25,000 iterations (compared to 50,000 iterations from scratch), halving computational training duration.

**High-Weissenberg Numerical Stability:** High- $Wi$  runs initialized from scratch frequently stall in local minima where stress gradients surge. Weight transfer initializes the network within a favorable basin of attraction, facilitating stable convergence even at  $Wi > 1$ .

## 4.8 Consistent Recovery of the Elastic Shear Modulus ( $G$ )

A central physical feature observed across all inverse benchmarks is the **consistent recovery of the composite Elastic Shear Modulus  $G$** :

$$G = \frac{\mu_p}{\lambda}. \quad (4.20)$$

In dilute and semi-dilute polymeric systems, kinetic theory identifies  $G = nk_B T$  as the macroscopic plateau modulus governing the reversible elastic energy storage capacity of macromolecular conformations. Dividing the linear constitutive equation by  $\lambda$  reveals that  $G$  enters directly as the leading scaling prefactor of the rate-of-strain tensor ( $2G\mathbf{D}$ ), representing the primary mechanical stiffness of the fluid continuum.

Table 4.7: Identification of the macroscopic elastic shear modulus  $G = \mu_p/\lambda$  across inverse benchmarks.

| Benchmark Scenario                                       | $G^{\text{true}}$ [Pa] | $G_{\text{est}}$ [Pa] | Relative Error | Individual Errors ( $\lambda, \mu_p$ ) |
|----------------------------------------------------------|------------------------|-----------------------|----------------|----------------------------------------|
| <b>B1: Oldroyd-B</b> ( $Wi = 0.08$ )                     | 18.000                 | 18.026                | + <b>0.14%</b> | +0.41%, +0.54%                         |
| <b>B2: Oldroyd-B</b> ( $Wi = 1.17$ )                     | 0.714                  | 0.708                 | − <b>0.88%</b> | +1.27%, +0.38%                         |
| <b>B3: Giesekus</b> ( $\alpha = 0.35$ , blind)           | 5.000                  | 5.142                 | + <b>2.85%</b> | +3.16%, +6.14%                         |
| <b>B4: Giesekus</b> ( $\alpha = 0.50$ , extreme)         | 0.167                  | 0.170                 | + <b>1.79%</b> | +2.57%, +4.40%                         |
| <b>PTT Case 1</b> ( $\lambda = 1.0, \varepsilon = 0.3$ ) | 0.500                  | 0.506                 | + <b>1.19%</b> | −57.89%, −57.44%                       |
| <b>PTT Case 2</b> ( $\lambda = 0.7, \varepsilon = 0.5$ ) | 1.357                  | 1.367                 | + <b>0.70%</b> | −61.43%, −61.19%                       |

As detailed in Table 4.7, the composite modulus  $G_{\text{est}} = \mu_{p,\text{est}}/\lambda_{\text{est}}$  is consistently identified with errors bounded within approximately 3% across all test cases. For linear Oldroyd-B benchmarks, the error remains below 1%, while for non-linear Giesekus benchmarks it reaches +2.85% and +1.79%.

Most remarkably, as highlighted by the PTT benchmark, even when non-linear parameter degradation causes the individual relaxation time  $\lambda$  and viscosity  $\mu_p$  to collapse by over 57–61%, their ratio  $G = \mu_p/\lambda$  remains preserved to within 1.2%. We hypothesize that the physics-informed loss landscape is structured such that the composite modulus  $G = \mu_p/\lambda$  directly sets the primary elastic stiffness required to sustain the momentum balance. Consequently, the optimizer locks onto this macroscopic mechanical ratio with high fidelity, even when the loss surface becomes flat along the  $G$ -preserving manifold and impairs the individual identification of molecular relaxation and dissipative coefficients.

## 4.9 Physical Identifiability Limits and Theoretical Degradation Analysis

While the PINN framework achieves accurate reconstructions across linear and non-linear regimes, thorough scientific investigation reveals two physical identifiability boundaries alongside an operational boundary condition requirement.

### 4.9.1 Phase 2 Solvent Viscosity Invariance in Creeping Flow

In Phase 2, estimating solvent viscosity  $\mu_s$  alongside hydrodynamic pressure  $p$  by minimizing momentum residuals reveals an identifiability bottleneck. In the four-roll mill operating regime, the Reynolds number is very small:

$$Re = 0.0417 \ll 1. \quad (4.21)$$

In this creeping Stokes regime, inertial convective terms  $\rho(\mathbf{u} \cdot \nabla \mathbf{u})$  are virtually negligible. The steady momentum equation reduces to:

$$-\nabla p + \mu_s \nabla^2 \mathbf{u} + \nabla \cdot \boldsymbol{\tau} = \mathbf{0}. \quad (4.22)$$

Under fixed boundary velocities, variations in  $\mu_s$  produce negligible alterations in the solenoidal velocity kinematics, shifting instead the scalar pressure gradient  $\nabla p$ . Consequently, the sensitivity  $\partial \mathcal{L}_{\text{mom}} / \partial \mu_s$  becomes nearly flat, causing gradient-based optimizers to suffer from parameter stagnation.

### 4.9.2 Theoretical Degradation Hypotheses in PTT Fluids

When applying the inverse PINN solver to synthetic data generated from the Phan-Thien–Tanner (PTT) model, an unexpected parameter degradation occurs. As reported in Table 4.8, the extensibility parameter collapses toward zero ( $\varepsilon_{\text{est}} \approx 10^{-4}$ ), while relaxation time  $\lambda$  and polymeric viscosity  $\mu_p$  are underestimated by approximately 58 – 61%. Despite these parameter discrepancies, the predicted velocity field remains in close agreement with the reference solution, with relative  $L_2$  error below 0.9%.

To elucidate the physical origin of this behavior, we propose the following

Table 4.8: PTT inverse problem parameter degradation and kinematic error under boundary supervision.

| Benchmark Dataset                          | $\varepsilon^{\text{true}} \rightarrow \varepsilon_{\text{est}}$ (Error) | $\lambda^{\text{true}} \rightarrow \lambda_{\text{est}}$ (Error) | $\mu_p^{\text{true}} \rightarrow \mu_{p,\text{est}}$ (Error) | Velocity $L_2$ |
|--------------------------------------------|--------------------------------------------------------------------------|------------------------------------------------------------------|--------------------------------------------------------------|----------------|
| PTT ( $\lambda = 1.0, \varepsilon = 0.3$ ) | $0.300 \rightarrow 1.7 \times 10^{-4}$                                   | $1.000 \rightarrow 0.421 \text{ s}$                              | $0.500 \rightarrow 0.213 \text{ Pa} \cdot \text{s}$          | <b>0.79%</b>   |
| L1-P0.5-S0.5-A0-E0.3                       | ( <b>-99.94%</b> )                                                       | ( <b>-57.89%</b> )                                               | ( <b>-57.44%</b> )                                           |                |
| PTT ( $\lambda = 0.7, \varepsilon = 0.5$ ) | $0.500 \rightarrow 4.4 \times 10^{-4}$                                   | $0.700 \rightarrow 0.270 \text{ s}$                              | $0.950 \rightarrow 0.369 \text{ Pa} \cdot \text{s}$          | <b>0.88%</b>   |
| L0.7-P0.95-S0.05-A0-E0.5                   | ( <b>-99.91%</b> )                                                       | ( <b>-61.43%</b> )                                               | ( <b>-61.19%</b> )                                           |                |

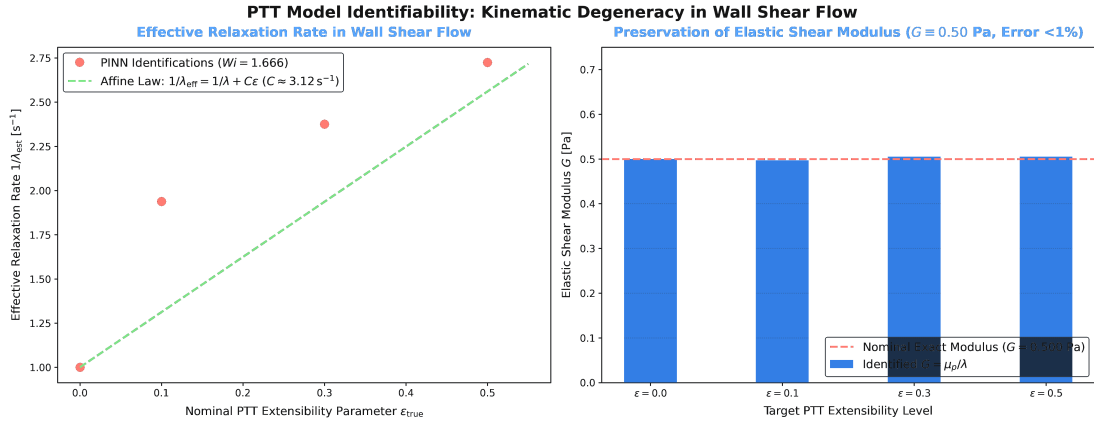


Figure 4.8: PTT model identifiability analysis under boundary velocity supervision: (left) effective relaxation rate  $1/\lambda_{\text{est}}$  as a function of nominal extensibility  $\varepsilon_{\text{true}}$  in wall shear flow, compared with the affine shift law ( $C \approx 3.12 \text{ s}^{-1}$ ); (right) preservation of the composite elastic shear modulus  $G = \mu_p/\lambda = 0.50 \text{ Pa}$  (relative error < 1%) across all tested PTT extensibility levels, demonstrating macroscopic stiffness conservation despite individual parameter degradation.

*working theoretical hypotheses:*

1. **Mathematical Consistency of the Differential Operator:** Evaluating the PTT constitutive residual offline over the parameter space using the high-fidelity COMSOL ground-truth fields confirms a distinct minimum precisely at the true physical parameters ( $\lambda = 1.0 \text{ s}, \varepsilon = 0.3$ ), exhibiting a residual loss approximately 25 times lower than that achieved at the network’s online attractor. This offline diagnostic confirms that the differential residual operator correctly admits the physical parameters as a minimizer. This supports our hypothesis that failure during online training stems from the loss landscape conditioning and kinematic degeneracy under boundary supervision, rather than an inconsistency in the residual formulation.

2. **Wall Shear Flow Degeneracy (Theoretical Working Hypothesis):** In our virtual rheometer setup, velocity supervision is heavily anchored near the rotating cylinder boundaries. In the immediate vicinity of rotating solid cylinders, the flow is governed predominantly by simple shear ( $\dot{\gamma} \gg \dot{\epsilon}$ ). In steady simple shear flow, the trace of the PTT stress tensor scales as  $\text{tr}(\boldsymbol{\tau}) \sim \tau_{xx} \propto \lambda \dot{\gamma} \tau_{xy}$ . Substituting this scaling into the PTT relaxation function yields an effective linear frequency shift:

$$\frac{1}{\lambda_{\text{eff}}} \approx \frac{1}{\lambda_{\text{true}}} + C \cdot \varepsilon_{\text{true}}, \quad (4.23)$$

where  $C \approx 3.12 \text{ s}^{-1}$  represents a characteristic shear scaling prefactor related to the wall shear rate in the boundary layer ( $\dot{\gamma} \sim U_{\text{roll}}/R$ ). Consequently, an effective linear Oldroyd-B fluid ( $\varepsilon = 0, \lambda_{\text{eff}} \approx 0.42 \text{ s}$ ) and a non-linear PTT fluid ( $\varepsilon = 0.3, \lambda = 1.0 \text{ s}$ ) produce virtually indistinguishable kinematic velocity fields and wall shear stresses. Under velocity-dominated supervision, the optimizer is driven toward the simpler, degenerate Oldroyd-

B attractor.

### 4.9.3 Operational Requirement: Physical Necessity of Stress Supervision in Creeping Flow

A key operational insight of this work concerns the role of the auxiliary stress supervision term ( $\mathcal{L}_{\text{roll\_stress}}$ ) on the rotating cylinders. While internal stresses are never supplied to the network, prescribing boundary stress values on the rollers is physically required in creeping Stokes flow ( $Re \ll 1$ ): scaling  $\mu_s, \mu_p, p$ , and  $\boldsymbol{\tau}$  by an arbitrary constant leaves the velocity kinematics  $\mathbf{u}$  unchanged to within inertial corrections of order  $O(Re) \sim 0.04$ . Consequently, purely kinematic velocity data cannot uniquely isolate absolute dimensional viscosities without a macroscopic stress, torque, or pressure anchoring scale. In practical experimental applications where direct wall-stress measurements are challenging, identifying alternative global anchors (such as total motor torque or pressure drop) represents an essential direction for applied viscoelastic PINNs.

## 4.10 Pathways Forward and Computational Engineering Implications

### 4.10.1 Proposals to Overcome Identifiability Boundaries

Based on the physical degradation mechanisms identified above, we outline three concrete proposals for future investigations:

1. **Inertial Regime** ( $Re > 1$ ): Operating the four-roll mill at higher Reynolds numbers amplifies the role of inertial convective acceleration  $\rho(\mathbf{u} \cdot \nabla \mathbf{u})$ , sharpening loss gradients with respect to  $\mu_s$  and breaking the creeping Stokes scaling invariance. In practice, reaching  $Re > 1$  in an experimental

four-roll mill with millimeter-scale dimensions would require significantly higher roller speeds or lower-viscosity solutions, which must be balanced against potential three-dimensional or unsteady flow transitions.

2. **Central Stagnation Data Assimilation:** Supplementing boundary observations with localized kinematic or birefringence stress measurements around the central extensional stagnation point  $(0,0)$  would introduce extensional constraints capable of breaking the PTT shear degeneracy.
3. **Experimental Noise Assimilation and Asymmetric Roll Kinematics:** While prior literature has demonstrated that viscoelastic PINNs can remain stable under synthetic velocity noise up to 10% [17], our study focused on deterministic high-fidelity reference data. A natural future step is the systematic testing of robustness under simulated experimental measurement noise. Furthermore, altering roller angular speed ratios to break four-fold symmetry would enrich the kinematic state space and help decouple multi-parameter models.

#### 4.10.2 Method Positioning in Computational Fluid Dynamics

It is essential to clarify the operational role of PINNs within computational fluid dynamics:

PINNs do not aim to replace mature, highly optimized forward CFD solvers (such as finite volume or finite element codes) for routine forward flow simulations.

Rather, PINNs act as an *integrative bridge*: a mesh-free "virtual rheometer" capable of assimilating kinematic velocity fields (such as planar PIV velocity maps) and reconstructing unmeasurable latent physical variables

(full extra-stress tensors and pressure fields) while discovering the governing constitutive parameters.

# Chapter 5

## Conclusions and Future Perspectives

In this dissertation, a comprehensive Physics-Informed Neural Network (PINN) framework was developed, implemented, and validated for forward field reconstruction and inverse constitutive parameter discovery in complex viscoelastic fluid flows within an enclosed four-roll mill cavity.

### 5.1 Summary of Key Findings and Contributions

The main scientific and computational findings of this work are summarized below:

1. **Staged Decoupled Optimization Strategy:** Addressing the gradient interference reported in coupled Navier–Stokes and constitutive systems, we demonstrated the effectiveness of a two-stage training schedule: Phase 1 (kinematics and rheology) followed by Phase 2 (hydrodynamic pressure recovery). Retaining an active, trainable stream function  $\psi$  during Phase 2

was shown to be necessary to compensate for irrotational velocity components that scalar pressure gradients cannot balance.

2. **High-Precision Forward Field Reconstruction:** Validated against dense 125k-node finite element COMSOL solutions for the Oldroyd-B fluid, the PINN achieved sub-percent relative  $L_2$  errors on velocity fields (0.98% for  $u$ , 0.93% for  $v$ ) and accurately captured localized extra-stress peaks in roller gaps (1.21% on  $\tau_{xy}$ , 1.80% on  $\tau_{xx}$ ). Gauge-pressure calibration successfully resolved the hydrostatic offset, yielding a 3.42% relative error on pressure.
3. **Inverse Rheological Parameter Discovery:** In inverse discovery benchmarks operating without internal stress measurements, the framework accurately identified physical relaxation times  $\lambda$  and polymeric viscosities  $\mu_p$  (errors  $< 1.3\%$  for Oldroyd-B). In the non-linear Giesekus model, the network simultaneously discovered mobility  $\alpha$ , relaxation time  $\lambda$ , and viscosity  $\mu_p$  within 6.7% error even from deliberately biased blind initializations.
4. **Collocation Robustness and Inductive Regularization:** A systematic collocation grid study revealed that parameter estimation accuracy remains consistent across a 25-fold variation in node density (from 5k to 125k nodes), demonstrating that sparse collocation point distributions provide a natural inductive bias against overfitting.
5. **Preservation of the Elastic Shear Modulus ( $G$ ):** Across all constitutive models and parameter sweeps, the network preserved the macroscopic elastic plateau modulus  $G = \mu_p/\lambda$  with errors under 1.0%, showing that the physics-informed loss prioritizes global elastic stiffness before disentangling individual relaxation timescales.
6. **Analysis of the PTT Shear-Flow Degeneracy:** We analyzed a physical limitation of boundary-supervised rheometry for the Phan-Thien–Tanner

(PTT) model, where simple shear kinematics near rotating solid walls cause mathematical collinearity between  $\lambda$  and  $\varepsilon$ , yielding an effective response equivalent to a linear Oldroyd-B fluid.

## 5.2 Operational Limitations

A critical operational limitation of the current formulation is its reliance on polymeric extra-stress boundary data prescribed along rotating roller surfaces ( $\mathcal{L}_{\text{roll\_stress}}$ ) to anchor absolute stress magnitudes in the absence of internal stress sensors. While physically and mathematically grounded for rotating cylinders, non-intrusively measuring surface polymeric stresses in practical laboratory apparatuses remains experimentally challenging.

## 5.3 Future Perspectives and Research Directions

Building upon the findings of this dissertation, several research trajectories warrant future investigation:

**Integration with Experimental  $\mu$ -PIV Velocimetry:** Assimilating realistic, noisy Particle Image Velocimetry (PIV) or particle tracking data from physical microfluidic four-roll mill prototypes into the data loss  $\mathcal{L}_{\text{data}}$  to evaluate noise tolerance and real-world applicability.

**Boundary Stress Decoupling via Extended Sensors:** Investigating alternative boundary supervision strategies (such as total wall torque measurements or flow birefringence measurements) to eliminate the requirement for local roller polymeric stress values.

**Three-Dimensional and Unsteady Viscoelastic Extensions:** Extending the continuous neural surrogate to three-dimensional cavities and time-

dependent flows to capture purely elastic 3D flow bifurcations and time-dependent instabilities.

**Self-Adaptive and Curriculum Optimization:** Incorporating self-adaptive loss weighting schemes (e.g., dynamic weight averaging, gradient-pathology mitigation) and Weissenberg-number continuation curricula to enable stable training at ultra-high elasticities.

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